HYPERTENSION



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BMCRI

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BASIC TERMS

Blood pressure: Force exerted by the blood against unit area of the vessel wall

Pulse Pressure: SBP-DBP

Mean arterial pressure: average of arterial pressures measured ms by ms over a period of time: (1/3 SBP + 2/3 DBP) or (DBP +1/3 PP)
Normal MAP: 70-110 mmHg

DEFINITION

Progressive cardiovascular syndrome characterised by the presence of blood pressure elevation to a level at which the **institution of therapy reduces blood pressure-related morbidity and mortality**

Office BP of **140/90 or higher** (ESH 2018)

Various guidelines define normal blood pressure differently.

EPIDEMIOLOGY

1) AGE AND SEX:

- **Increases with age**: tracks over time
- **Higher in males** till **middle age**
- DBP increases till 55 and then falls: **pulse pressure increases** henceforth.

2) RACE:

- Associated with **environmental factors**

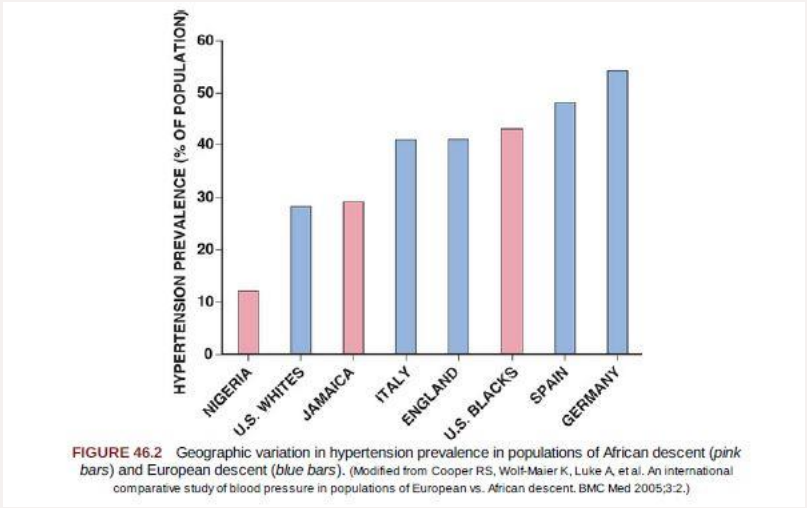
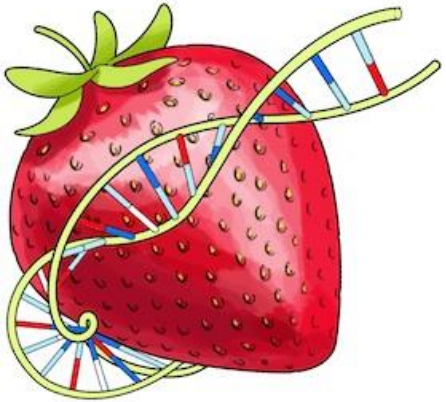


FIGURE 46.2 Geographic variation in hypertension prevalence in populations of African descent (*pink bars*) and European descent (*blue bars*). (Modified from Cooper RS, Wolf-Maier K, Luke A, et al. An international comparative study of blood pressure in populations of European vs. African descent. BMC Med 2005;3:2.)



3) ENVIRONMENTAL FACTORS:

- **Obesity**
- **Smoking and alcohol** intake
- **High dietary NaCl** and **low dietary calcium and potassium** intake
- Psychosocial stress
- **Poor physical activity**
- **Caffeine** but not coffee

4) GENETIC FACTORS:

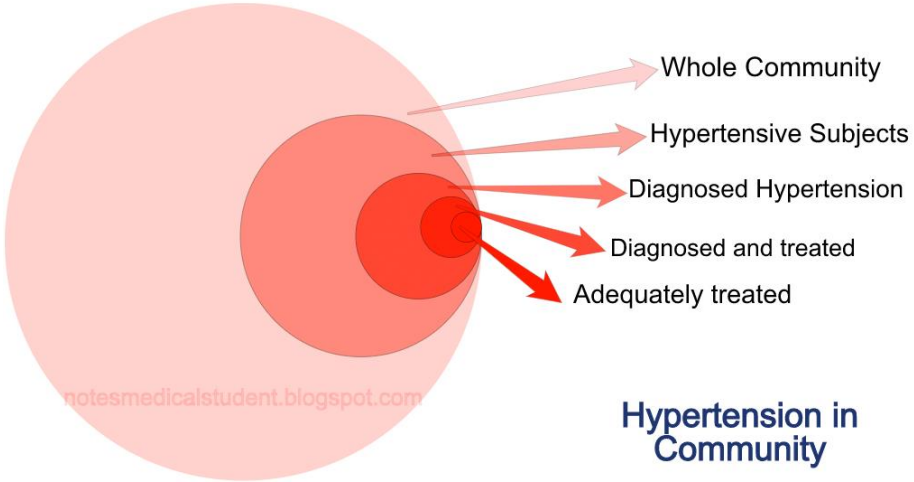
- Heritable (30-50%)
- **Polygenic** disorder; **Epigenetic** processes involved
- Few rare Mendelian forms (<2%)
- May also determine **target organ damage** and **response to therapy**

Magnitude of the problem

Commonest cardiovascular disorder

ICEBERG disease: “rule of halves”

Only half of the hypertensives in the general population are aware of the condition; only half of those aware are being treated; and only half of those treated are adequately treated



Magnitude of the problem

But, according to NHANES data in 2007-2010: **81% were aware** of hypertension; 74% were being treated but only **52% were controlled**.

Globally, the overall prevalence in adults aged 25 years and over was around **40 %** in 2008.

India:
Prevalence of hypertension was varying from **17 to 21 %** in all the states with marginal rural-urban differences. (2007-8 ICMR)

IGH –IV:
Overall prevalence: **29.8 %** Urban area: 33.8 % and rural: 27.6;
Control level: 20% in urban and 11% in rural population

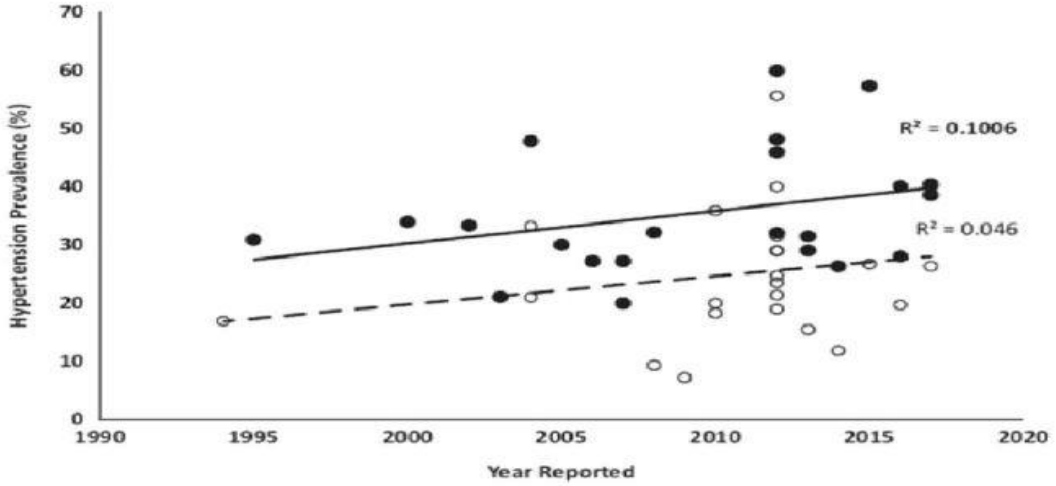


Fig. 1 Prevalence of hypertension in India over last four decades. Increasing trend in hypertension prevalence in India in urban (top panel) and rural (bottom panel) populations according to cross sectional regional studies from 1990s to date.

Table 5 Increasing trends in deaths and disability adjusted life years (DALYs) due to high systolic blood pressure in India (Global Burden of Diseases Study 2016).

	1990	1995	2000	2005	2010	2016
Deaths absolute numbers (thousands)	784.7	885.3	1005.0	1153.6	1385.6	1634.7
Death rate/100,000	90.8	92.7	95.9	101.3	113.1	124.2
% of total deaths	8.9	9.9	10.8	12.2	14.4	16.7
DALY's absolute numbers (millions)	20.9	23.4	26.2	29.3	34.4	39.4
DALY rate/100,000	2415	2451	2497	2576	2807	3000
% of total DALYs	3.9	4.4	5.0	5.7	7.0	8.5

CLASSIFICATION

PRIMARY HYPERTENSION (90-95%)

No specific underlying cause

Hemodynamic subtypes:

- Isolated systolic in young
- Isolated diastolic in middle age
- Isolated systolic in old

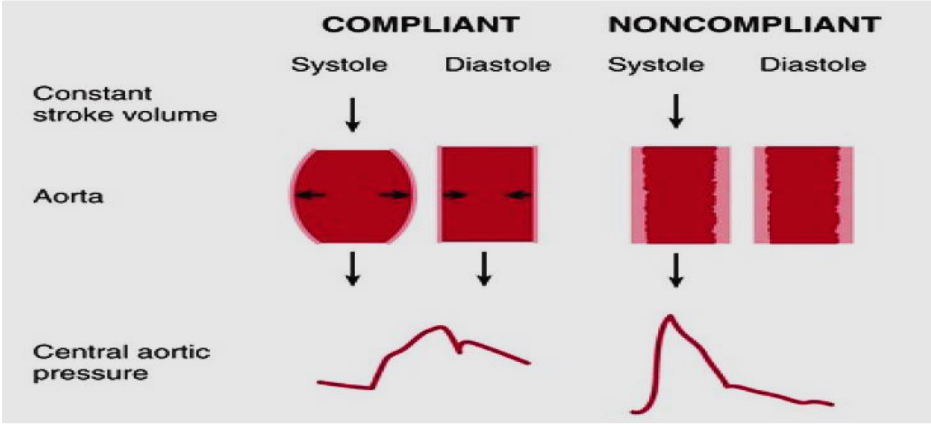
SECONDARY HYPERTENSION

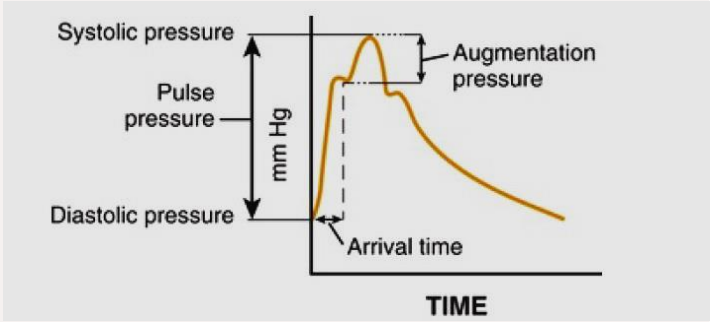
Has a specific underlying cause

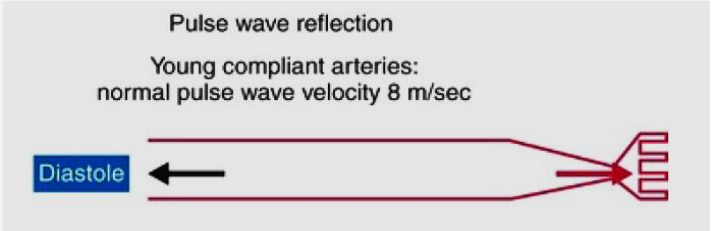
ISOLATED SYSTOLIC HYPERTENSION IN OLD

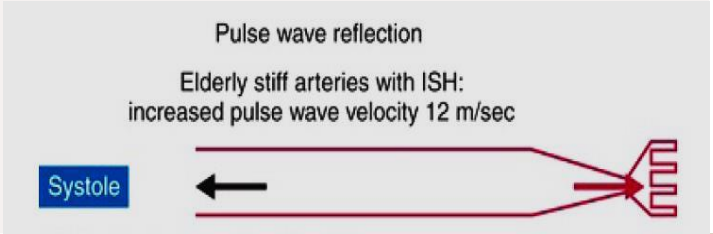
After 55 years DBP falls
SBP>140mmHg and DBP <90mmHg: **Wide PP**

Stiff central aorta (raised **collagen to elastin** ratio)
Rapid reflection of pulse waves from the periphery, **augmenting systolic aortic pressure**

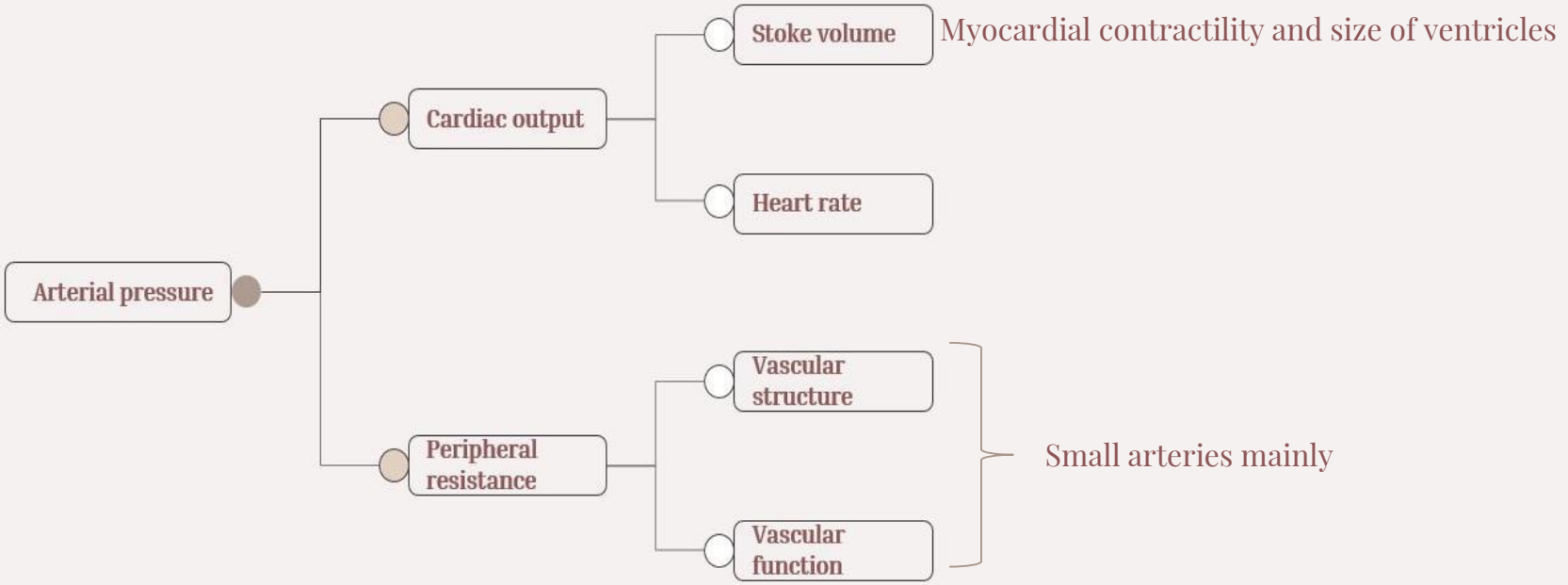








ETIOLOGY AND PATHOPHYSIOLOGY



Determinants of blood pressure

1) INTRAVASCULAR VOLUME

$$\text{Blood Flow} = \frac{\text{pressure across the vascular bed}}{\text{vascular resistance}}$$

PHYSIOLOGICAL:
Sodium: Main ECF Ion: determines ECF volume
When **increased**, kidneys **excrete Na and H20** to **maintain normal volume and blood pressure**

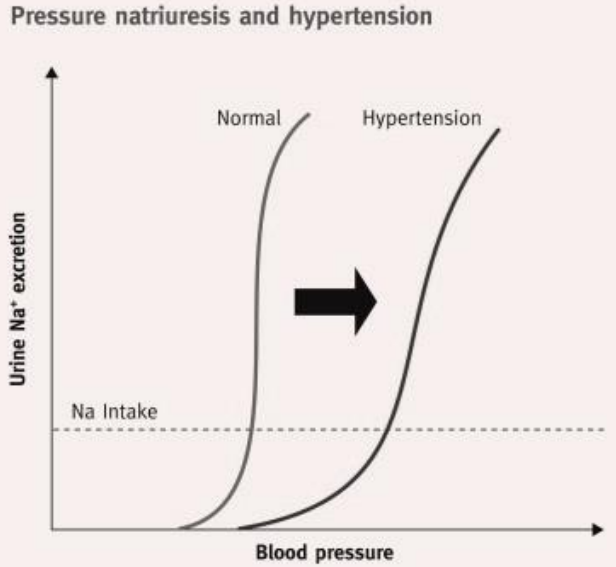
IN HYPERTENSION:
Kidneys **fail to excrete Na and water**:
i) **NaCl intake exceeds kidneys ability** (or)
ii) **Renal disease**
iii) Inappropriate **high mineralocorticoid action**
Initially high BP relates to ⬆ cardiac output
Later vascular resistance increases to maintain **constant blood flow** to organs.
Pressure natriuresis curve shifts to right

Pressure natriuresis curve reset:

Sodium balance is **maintained at the expense of raised arterial pressure.**

Profound nocturia in pure autonomic failure (supine **nocturnal hypertension**)

Nocturia may be a symptom of **uncontrolled primary HTN**



From Smith T, Pinnock C, Lin T, Jones R. *Fundamentals of Anaesthesia*, 3rd Edn, 2009. With kind permission of Cambridge University Press.

2) NEURAL MECHANISMS

PHYSIOLOGICAL:
Short term: adrenergic reflexes
Long term: adrenergic function

Baroreceptor reflex: stretch sensitive
Carotid sinus and aortic arch
Firing increases with BP and decreases SNS outflow: HR and BP
,

IN HYPERTENSION:
Associated with **increased SNS outflow**
More so in **obesity related HTN** and **OSA**

Baroreceptor **adapts to sustained rise (reset)**

Autonomic neuropathy: labile BP with **difficult to control BP spikes**
Partial dysfunction: Orthostatic hypotension, supine hypertension and symptomatic postprandial hypotension

SNS: sympathetic nervous system

3) HORMONAL MECHANISM: RENIN-ANGIOTENSIN-ALDOSTERONE

PHYSIOLOGICAL:
Vasoconstrictor properties of **angiotensin II**
Sodium retaining property of **aldosterone**

Renin is released from **JG cells** i/r/t:

- **Decreased NaCl** in ascending limb of LOH near the afferent arteriole
- Decreased stretch of afferent arteriole **(low volume)**
- **SNS stimulation** via β1 receptors

Inhibited by:

- **High NaCl,**
- **High volume,**
- **Angiotensin II**

IN HYPERTENSION:

10-15% have high plasma renin activity
In high sodium states even some RAAS activity is inappropriate
Endothelial dysfunction and vascular remodeling (amplified by salt intake)

Secondary HTN:
Renin secreting tumors: hemangiopericytomas of JG apparatus; rarely RCC and other carcinomas
Renal artery stenosis

Prorenin: inactive precursor: increased TGF-β production: **fibrosis**

Angiotensin II acts via **GPCR receptors**

AT1: pressor; aldosterone release; **atherosclerosis; cardiac hypertrophy; insulin resistance, renal damage**

AT2: vasodilation, sodium excretion, inhibits cell growth and matrix formation

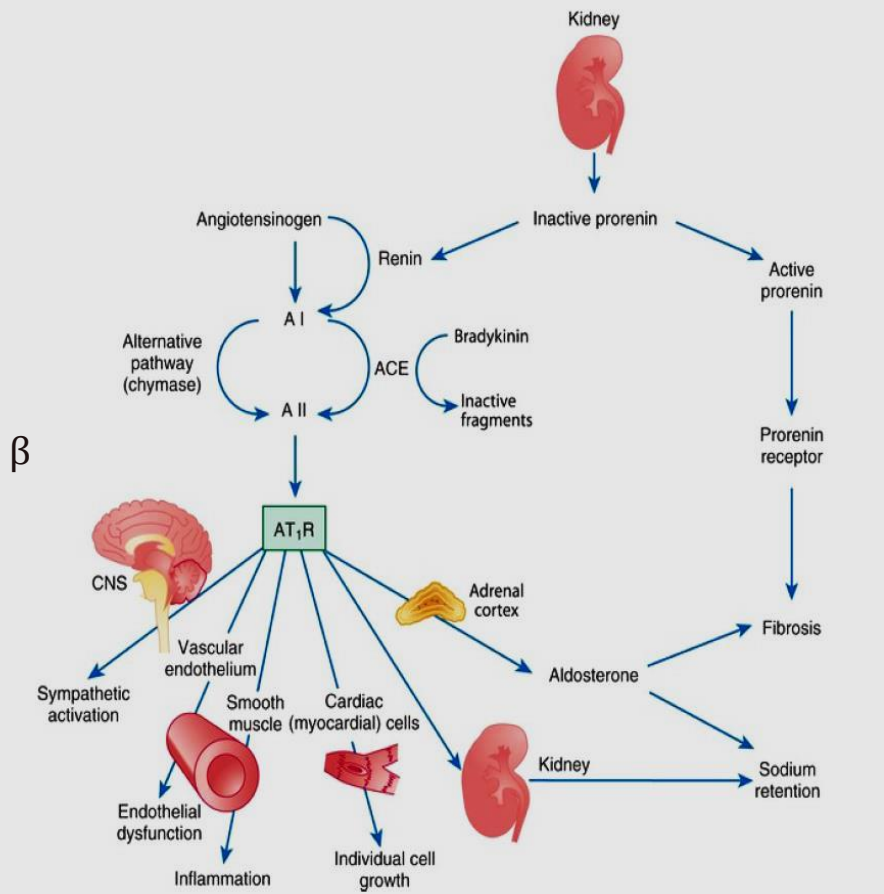
Aldosterone:

Secreted by Zona Glomerulosa i/r/t

- Angiotensin II
- High Potassium states
- ACTH (mild)

Acts via ENaC channel: Na and H20 absorption; K and H loss

MC excess leads to: **myocardial fibrosis, LVH, nephrosclerosis, vascular remodeling; SNS activity (via brain stem receptors)**



4) VASCULAR MECHANISMS

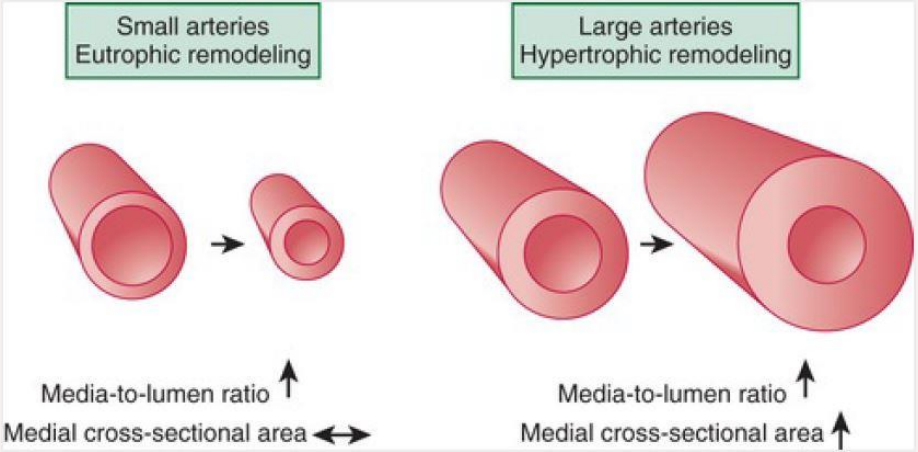
Resistance is **inversely proportional** to the fourth power of **radius**

Vascular remodeling: eu/hypertrophic
Increased media/lumen ratio

Stiff rigid vessels: pressure increases more with volume (less buffering): cause and effect

Intracellular pH: tone and growth of smooth muscles
Na-H exchange: increased in HTN:
i) Increased Na entry: Ca entry: **increased tone**
ii)Increased pH: enhances Ca sensitivity

Endothelial cell dysfunction



SMCs rearrange themselves around a smaller lumen diameter

Increased size of SMCs and accumulation of EMP (collagen) due (TGF)-β.

Endothelial cell dysfunction:

Normal endothelium secretes: (i/r/t **ACh, bradykinin,** sheer stress)
NO gas: potent vasodilator
Endothelium derived hyperpolarizing factor
Impaired in HTN
Evidence of **improvement** with **aerobic exercise, weight loss and antihypertensives**.

Smoldering **vascular inflammation**:
Cause and effect: as suggested by **elevated CRP**

Oxidative stress
ROS(mainly **superoxide**) **quench NO**
Sources: NADPH oxidases; NOS uncoupling: Xanthine oxidase

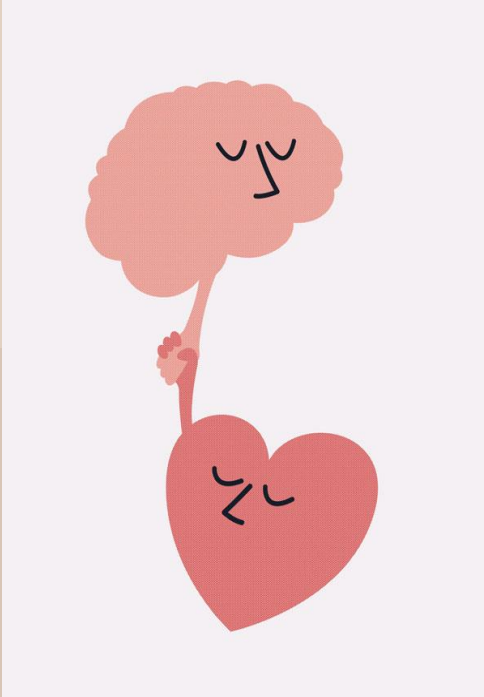
Pathological consequenses

In hypertensive individuals, cardiovascular risk increases sharply with:

BP stage

Additional risk factors

Target-organ damage



1) HEART



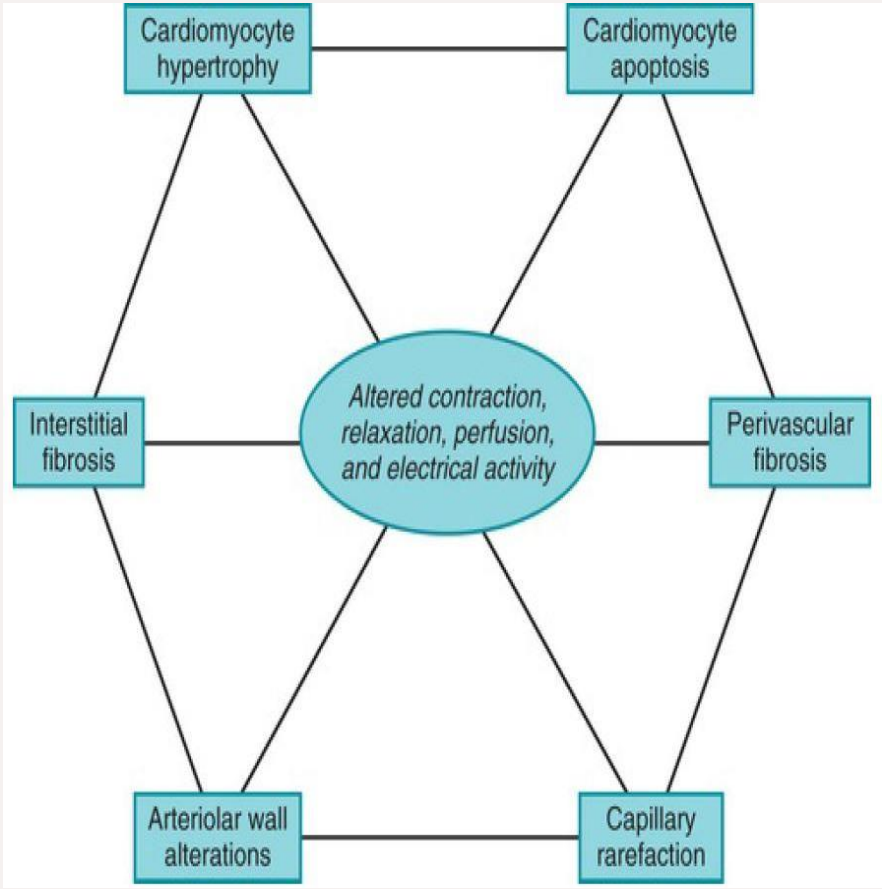
Most common cause of death in hypertensives
HTN accounts for **50% of IHD**

Adaptations of heart: **LVH :** represents **failure to diagnose and treat hypertension adequately**
CHF atherosclerotic CAD, microvascular disease and arrhythmia

Pathogenesis of hypertensive heart disease:

- Pressure overload
- Neurohumoral activation
- Inflammation
- Fetal reprogramming of cardiomyocyte genes

Accelerated by:
A II, aldosterone, NE, prorenin



(medial hypertrophy)

Classification of Hypertensive Heart Disease

Class I: Subclinical diastolic dysfunction by echocardiography without left ventricular hypertrophy
Asymptomatic patients with abnormal left ventricular relaxation/stiffness by Doppler echocardiography, a common finding in hypertensive individuals >65 years
Class II: Left ventricular hypertrophy
IIA: with normal functional capacity (NYHA Class I)
IIB: with abnormal functional capacity (NYHA Class >II)
Class III: Heart failure with preserved ejection fraction (HFpEF)
Class IV: Heart failure with reduced ejection fraction (HFrEF)

NYHA, New York Heart Association.





Diastolic dysfunction (1/3 of patients with CHF):
Early consequence
LVH and ischemia augments

Mechanism:
Normal resting coronary blood flow, but **vasodilator reserve impaired**
Myocardial microvasculature doesn't grow proportional to muscle mass (**capillary rarefaction**)
Microvascular ischemia
Even in the **absence of atherosclerosis**: **subendocardial ischemia** when oxygen demand increases.
Subendocardial ischemia and cardiac fibrosis: impairs diastolic relaxation (HFpEF)

2) BRAIN:



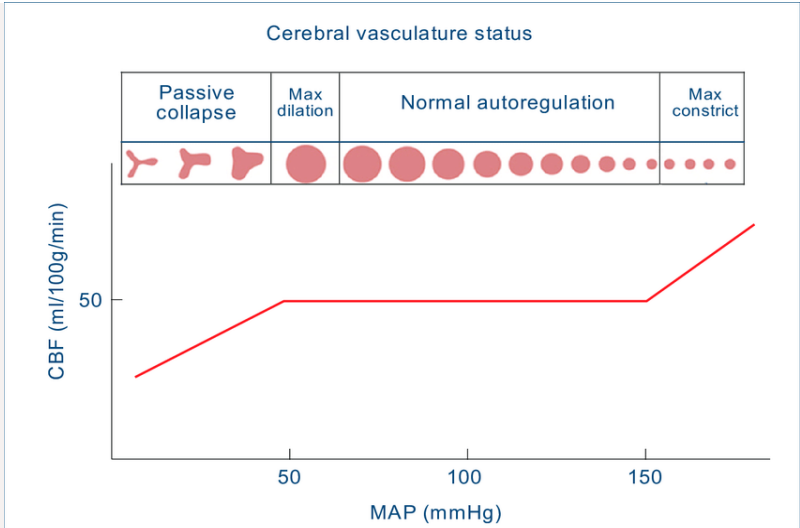
SROKE: 2ND most common cause of death worldwide
Hypertension: accounts for 50-66% of stokes
Incidence raises progressively (SBP; >65 years)

IMPAIRED COGNITION:
Beta amyloid deposition: **dementia**
Either **single large vessel or multiple small vessel disease**

CEREBRAL BLOOD FLOW AUTOREGULATION:

BF remains unchanged over **MAP of 50-150**
Shifts to higher blood pressure zone in HTN

Failure: Malignant hypertension leads to **encephalopathy**



3) KIDNEY:



Cause and target organ; 2nd to DM in cause for CKD

Renal damage: graded continuous and present throughout the distribution of BP above optimal
SBP>DBP

Hypertensive nephrosclerosis
Preglomerular arterioles affected: **Ischemia of glomeruli and post glomerular structures**

Vicious cycle:
Loss of autoregulation: high pressures reach glomerulus: hyper-filtration, hypertrophy, FSGS

Glomerular sclerosis and **ischemic atrophic renal tubules** : **small scarred kidney**
In **malignant HTN: fibrinoid necrosis** of the afferent arterioles, rarely focal necrosis of the glomerular tuft

Microalbuminuria: **early sensitive marker of CKD** and **predicts CVD risk.**

4) PERIPHERAL ARTERIES:



Cause and target organ

Vascular disease:
Hyaline arteriosclerosis
Major contributor to **stroke, heart disease and renal failure**

PVD: intermittent claudication: (ABPI <0.9)

Aortic dissection: overwhelming majority

Abdominal aortic aneurysm:
USG screening recommended in hypertensives and smokers over 65 years
Compulsory if pulsations are detected below umbilicus (most occur below origin of renal arteries)

5) RETINA: Hypertensive retinopathy

Grade	Stage	Description	Ocular symptoms	Systemic involvement
1	Mild hypertension	Mild generalised arteriolar narrowing or sclerosis	No symptoms	None
2	More marked hypertension	Definite focal narrowing and arteriovenous crossings. Moderate to marked sclerosis of the retinal arterioles. Exaggerated arterial light reflex	No symptoms	None or minimal
3	Mild angiospatic retinopathy	Retinal haemorrhages, exudates and cotton wool spots. Sclerosis and spastic lesions of retinal arterioles	May be symptomatic	Cardiac, cerebral or renal dysfunction are frequently identifiable
4	Severe hypertensive retinopathy	Severe Grade III signs and papilledema. Elschning's spots are present in some	Symptomatic	Cardiac, cerebral and renal dysfunction are more severe

The first classification of hypertensive retinopathy, proposed in 1939 by Keith, Wagener and Barker, based on the progressive severity of ophthalmoscopic signs.

First Bibliographic Reference: Keith NM, Wagener HP, Barker NW. Some different types of essential hypertension: their course and their prognosis. Am J Med Sci 1939;197:332-43.

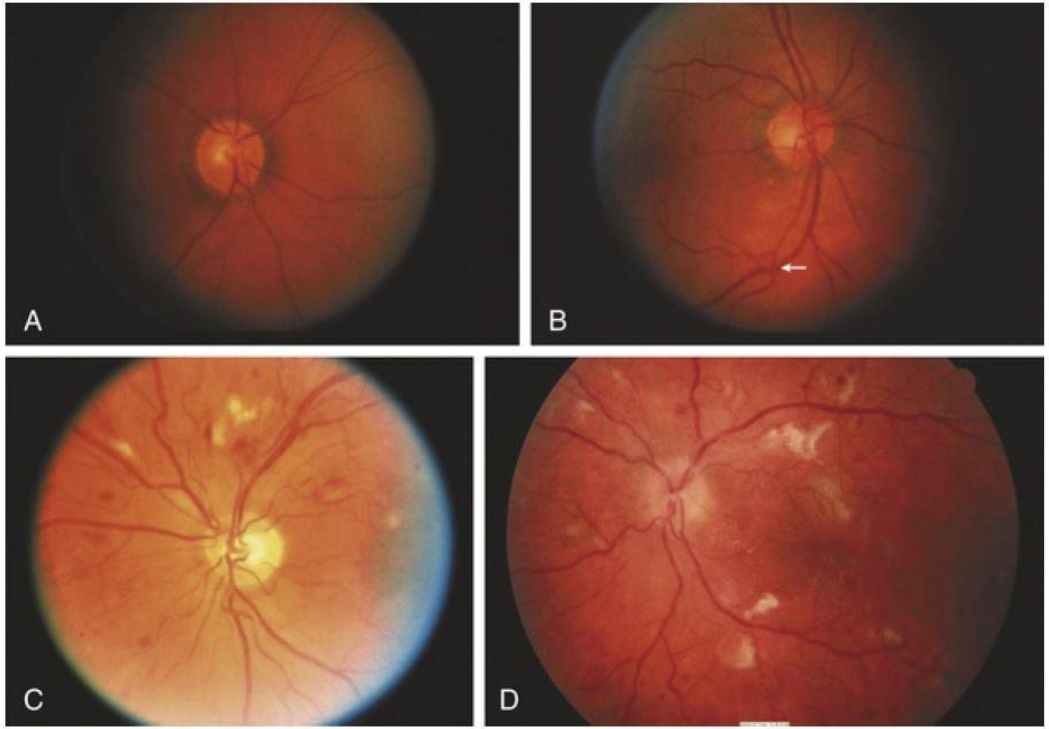


FIGURE 47.10 Retinal photographs showing the stages of hypertensive retinopathy. **A,** Mild diffuse arteriolar narrowing. **B,** Arteriovenous nicking *(arrow).* **C,** Hemorrhages and exudates. **D,** Papilledema. (From Grosso A, Veglio F, Porta M, et al. Hypertensive retinopathy revisited: some answers, more questions. Br J Ophthalmol 2005;89:1646.)

Risks Influencing Prognosis in Patients with Hypertension

Risk Factors for Cardiovascular Disease
Systolic and diastolic BP levels
Levels of pulse pressure (in elderly patients)
Age: men >55 years; women >65 years
Smoking
Dyslipidemia (LDL-C >115 mg/dL)
Impaired fasting glucose (102-125 mg/dL) or abnormal glucose tolerance test result
Family history of premature cardiovascular disease
Abdominal obesity
Diabetes mellitus
Subclinical Target Organ Damage
Left ventricular hypertrophy
Carotid wall thickening or plaque
Low estimated glomerular filtration rate ≤60 mL/min/1.73 m^2
Microalbuminuria
Ankle-brachial BP index <0.9
Established Target Organ Damage
Cerebrovascular disease: ischemic stroke, cerebral hemorrhage, transient ischemic attack
Heart disease: myocardial infarction, angina, coronary revascularization, heart failure
Renal disease: diabetic nephropathy, renal impairment
Peripheral arterial disease
Advanced retinopathy: hemorrhages or exudates, papilledema

Secondary Hypertension

Suspected in:

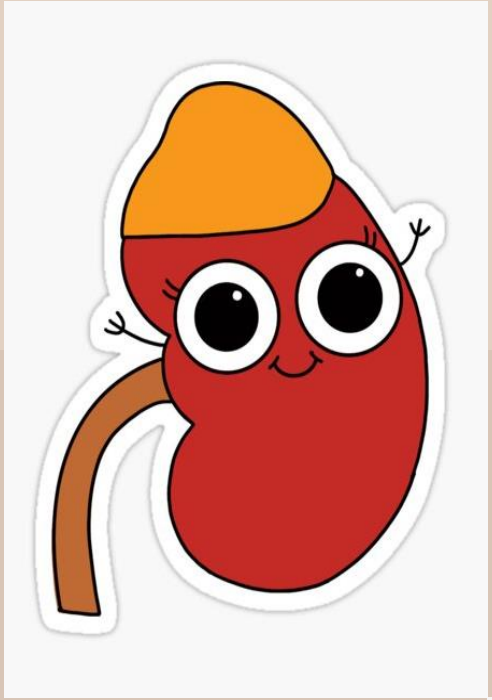
Young hypertensive

Resistant hypertension

Target organ damage disproportional to duration/severity of HTN

Predominant diastolic

Unprovoked hypokalemia with HTN



1) RENAL PARENCHYMAL DISEASE

Most common cause of secondary hypertension (2-5%)
Most common cause of CKD: DM and HTN (>chronic glomerulonephritis)
HTN is present in >80 patients with CKD
2/3 CKD patients: **nocturnal masked HTN**
More severe in glomerular than interstitial diseases

To figure out the primary disease:

- Proteinuria>1g/d
- Active urine sediment

Indicate primary renal disease

Goal of treatment: control BP and retard progression of renal disease
Most need: ACE inhibitor/ARB + Diuretic + CCB
Overly aggressive treatment: AKI

ACUTE RENAL DISEASES:

Sudden insult causing:

- Impaired Na and water excretion (volume expansion), or
- Reduced renal blood flow (e.g., b/l renal ischemia because of cholesterol emboli), or
- RAAS activation.

Hypertension can be the **presenting sign of systemic vasculitis** involving the kidney.

Drugs that can precipitate **AKI in CKD**:

- **NSAIDS**: reduce PGs (vasodilators within the kidney)
- **ACE inhibitors and ARBS**: ARF in b/l renovascular disease (perfusion depends on AG II)

POST RENAL TRANSPLANT:

Stenosis at site of anastomosis/ high dose **steroids**
Native kidney can secrete **renin**: ACE inhibitors/ARBs

2) RENOVASCULAR DISEASE

Potentially curable

Spectrum of disease:
(1) Incidental renal artery stenosis
(2) Renovascular hypertension with reduced perfusion (RAAS activation);
(3) Accelerated cerebrovascular disease (diastolic dysfunction, heart failure, and stroke)
(4) Ischemic nephropathy with renal tissue hypoxia, extensive microvascular disease, and progressive renal atrophy

Classification:

(1) Atherosclerotic renal artery stenosis (85%): origin of renal artery, older patients
(2) Fibromuscular disease: distal 2/3 and branches of the renal arteries; women(20-60 years)
(3) Emboli in renal arteries
(4) Extrinsic compression of renal arteries



MECHANISM:

Reduced renal blood flow: **activates RAAS**

Initially **contralateral kidney compensates**: normal blood volume and pressure

Later **contralateral kidney becomes damaged**: **volume expansion** and **hypertension**

Renin secretion from the stenotic kidney **decreases**

Eventually: **ischemia** leads to **microvascular disease, renal atrophy, and fibrosis.**

DIAGNOSIS:

Clinical Clues for Renovascular Hypertension

1. Onset of hypertension before age 30 or after 50
2. Acceleration of treated primary hypertension
3. Deterioration in renal function in treated primary hypertension
4. Acute kidney injury (AKI) during treatment of hypertension
5. Flash pulmonary edema
6. Progressive renal failure
7. Refractory heart failure
8. Unilateral small (atrophic) kidney size by ultrasound examination

ARAS: Triad of drug refractory hypertension, flash pulmonary edema, and ischemic nephropathy
FMD: Young to middle-age women little or no family history of hypertension
A whooshing or swishing sound with each heartbeat: carotid FMD

MANAGEMENT:

Screening test: **Duplex USG**, (75% sensitive and 90% specific) (or) **OIH scan** (or) captopril DTPA
Confirmation: Spiral CT or **MR angiography** (beware in advanced CKD) or **DSA**

FMD:
Rx of choice: **Balloon angioplasty without stenting** (excellent outcome)

ARAS:
Conservative approach: Control cardiovascular risk factors
Smoking cessation; statins and antiplatelet drugs
ACEI- or ARB for HTN

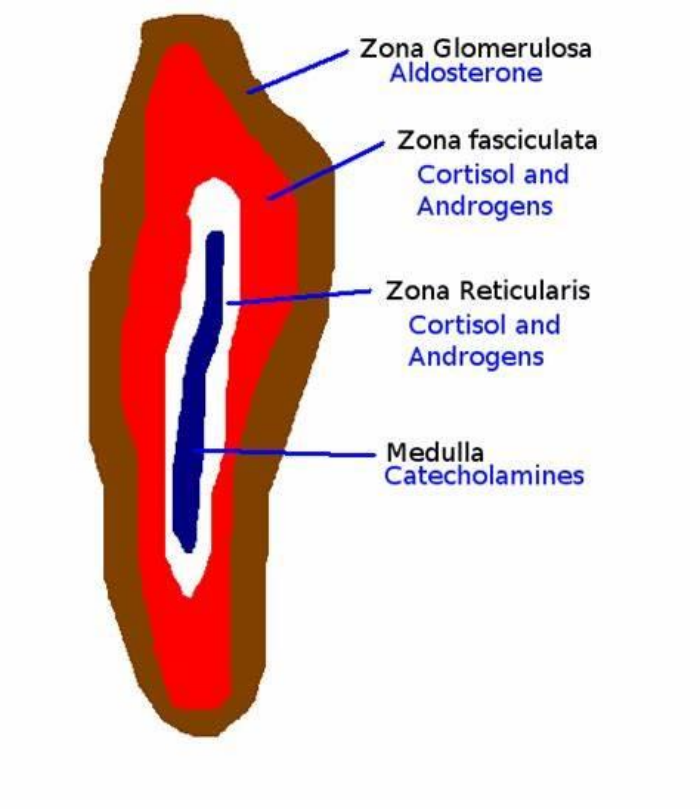
Indications for stenting:
(1) Intractable hypertension,
(2) AKI induced by ACEI or ARB
(3) Recurrent, episodic (flash) pulmonary edema.

3) ADRENAL CAUSES

Less than 1% of all cases:

Includes

- Primary hyperaldosteronism
- Cushing's syndrome
- Pheochromocytoma



PRIMARY HYPERALDOSTERONISM

Aldosterone independent of RAAS system: **low PRA**
Most common cause of mineralocorticoid excess
Potentially curable

3rd–5th decade
Refractory HTN / HTN with unprovoked hypokalemia / K<3.1 on diuretics
Asymptomatic > polyuria, polydipsia, paresthesias, muscle weakness (hypokalemic alkalosis)
No edema

Unilateral adenoma or bilateral hyperplasia (> 90%)

Pathogenesis of adenoma:
K-channel mutations: abnormally permeable to Na: depolarization and Ca entry: MC secretion and cell proliferation

DIAGNOSIS:

Screening:
PA: >15 ng/dL and PRA <1.0
PA/PRA > 30:1 (+ aldosterone concentration >20 ng/dL)
W/H aldosterone antagonists: 4-6 weeks; hypokalemia correction

Confirmation:
Failure of aldosterone suppression after:
Oral 3 day NaCl load (recommended) or,
2L NS infusion over 4 hours or,
fludrocortisone or captopril

Localization:
Adrenal vein sampling > CT
CT can miss microscopic adenomas and benign incidentomas are common
Adrenal scintigraphy after dexamethasone suppression

MANAGEMENT:

Unilateral adenoma:
Laparoscopic adrenalectomy
Curative in 40-70%
May cause transient hypoaldosteronism

Bilateral hyperplasia:
Medically:
Salt restriction
Aldosterone antagonists: Spironolactone/eplerenone
Add on: Potassium sparing diuretics

DIFFERENTIAL DIAGNOSIS: (severe HTN and hypokalemia)

Rare Mendelian forms of MC excess:
Premature onset (< 30),
Severe hypertension
Family history

1) Liddle's syndrome:
Mutation activating ENaC channel
Rx: Amiloride/Triamterene

2) Familial Glucocorticoid-remediable Aldosteronism

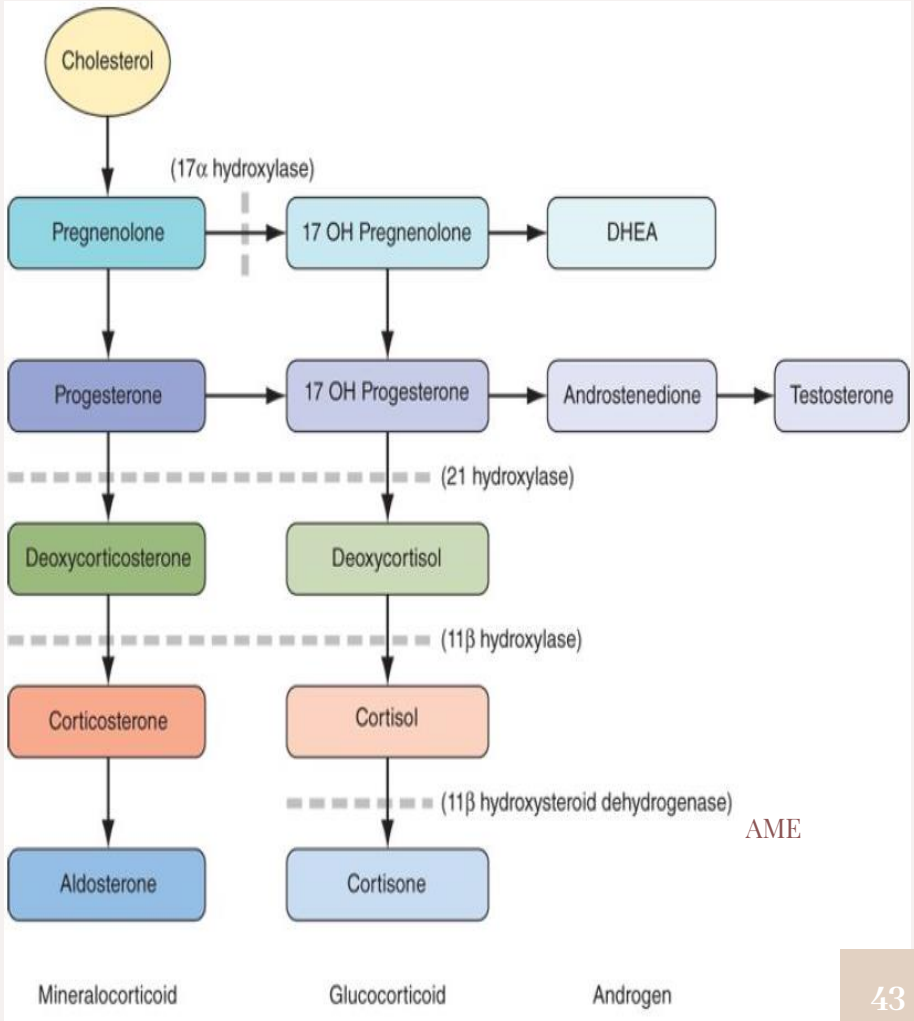
3) Apparent mineralocorticoid excess

4) Congenital Adrenal hyperplasia

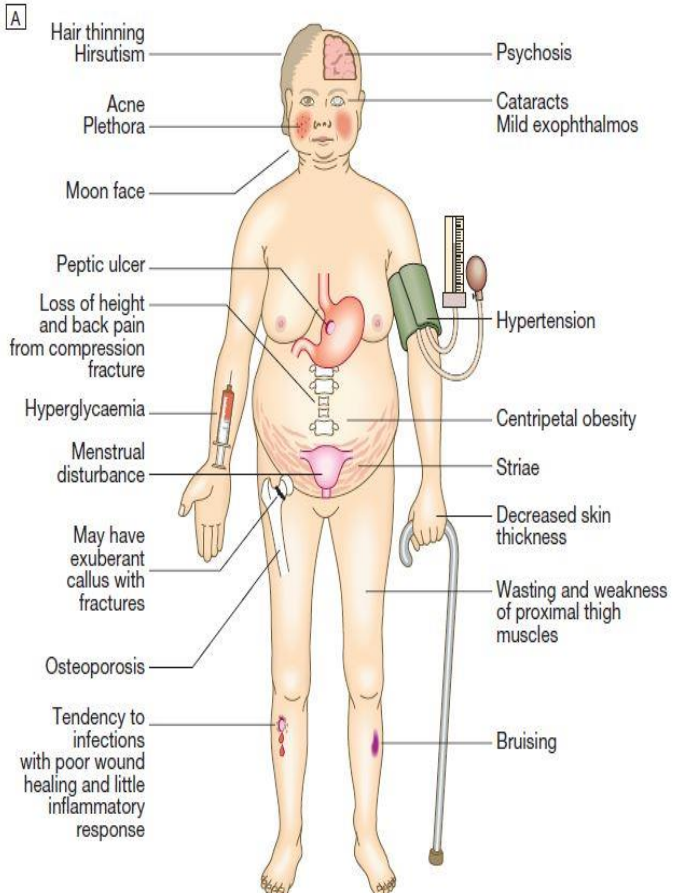
4) Congenital Adrenal hyperplasia

i) 17α-hydroxylase deficiency:
Glucocorticoids and androgens low
Pseudo hermaphroditism in male
Absent SSS in females
Excess ACTH (loss of neg feedback): stimulates cortex
Excess MC (mainly deoxycorticosterone)
Rx: low dose glucocorticoids

ii) 11β-hydroxylase deficiency:
Most common
Low GC, high MC and androgens
Presentation:
Newborn: ambiguous genitalia/penile enlargement
Childhood: precocious puberty and short stature.
Older: Hirsutism, menstrual irregularities



OTHER ADRENAL CAUSES



Cushing's syndrome
75-80% have hypertension
Endo/exogenous steroids
Excess cortisol
Can exceed capacity of conversion to cortisone
90% ACTH dependent

Management based on cause

Pheochromocytoma (and paragangliomas) (0.05% of HTN)
20%: MEN 2A/2B; VHL, NF1
Triad:
Episodic headache, profuse sweating, tachycardia
Orthostatic hypotension

Definitive management:
Surgical excision
Medical prep before surgery

4) MISCELLANEOUS CAUSES

COARCTATION OF AORTA:

Most common congenital cardiovascular cause of HTN
35% of children with Turner's syndrome

Radio-femoral delay and **Radio-radial delay** +/-
Systolic murmur in left posterior ISA

Diagnosis: X-ray or Transesophageal ECHO or cardiac MRI

Management: surgical repair or balloon angioplasty

4) MISCELLANEOUS CAUSES

ENDOCRINE:

- Acromegaly
- Hypothyroidsm: Mild diastolic
- Hyperthyroidsm: Systolic
- Hyperparathyroidsm (and hypercalcemia of any etiology)

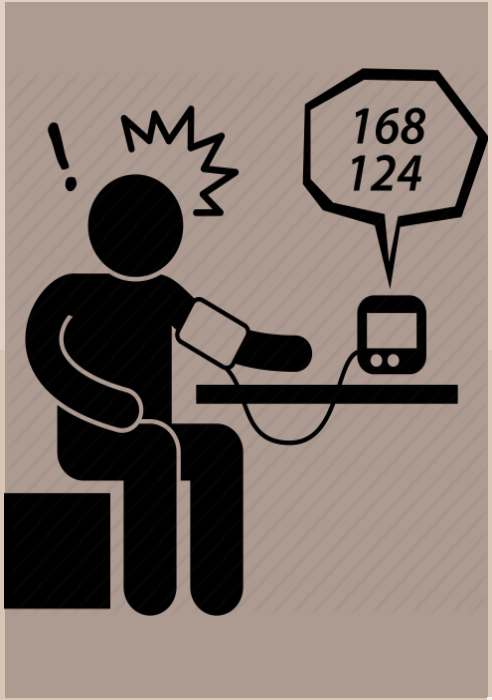
SYSTOLIC HTN WITH WIDE PULSE PRESSURE:

- AR
- Thyrotoxicosis
- Fever
- AV Fistula
- PDA

Renal	Parenchymal diseases, renal cysts (including polycystic kidney disease), renal tumors (including renin-secreting tumors), obstructive uropathy
Renovascular	Arteriosclerotic, fibromuscular dysplasia
Adrenal	Primary aldosteronism, Cushing's syndrome, 17α-hydroxylase deficiency, 11β-hydroxylase deficiency, 11-hydroxysteroid dehydrogenase deficiency (licorice), pheochromocytoma
Aortic coarctation	
Obstructive sleep apnea	
Preeclampsia/eclampsia	
Neurogenic	Psychogenic, diencephalic syndrome, familial dysautonomia, polyneuritis (acute porphyria, lead poisoning), acute increased intracranial pressure, acute spinal cord section
Miscellaneous endocrine	Hypothyroidism, hyperthyroidism, hypercalcemia, acromegaly
Medications	High-dose estrogens, adrenal steroids, decongestants, appetite suppressants, cyclosporine, tricyclic antidepressants, monoamine oxidase inhibitors, erythropoietin, nonsteroidal anti-inflammatory agents, cocaine
Mendelian forms of hypertension	

Evaluation of a hypertensive patient

History and GPE
Screen for:
CVD risk factors
Secondary causes
Target organ damage



History

Duration of hypertension

Previous therapies: responses and side effects

Family history of hypertension and cardiovascular disease

Dietary and psychosocial history

Other risk factors: weight change, dyslipidemia, smoking, diabetes, physical inactivity

Evidence of secondary hypertension: history of renal disease; change in appearance; muscle weakness; spells of sweating, palpitations, tremor; erratic sleep, snoring, daytime somnolence; symptoms of hypo- or hyperthyroidism; use of agents that may increase blood pressure

Evidence of target organ damage: history of TIA, stroke, transient blindness; angina, myocardial infarction, congestive heart failure; sexual function

Other comorbidities

Physical Examination:

- Body habitus
- Blood pressure in both arms
- Supine and standing blood pressures
- Funduscopic examination of retina
- Vascular and abdominal bruits
- Cardiac rate and rhythm
- Signs of congestive heart failure
- Signs of secondary hypertension

Lab Investigations:

System	Test
Renal	Microscopic urinalysis, albumin excretion, serum BUN and/or creatinine
Endocrine	Serum sodium, potassium, calcium, TSH
Metabolic	Fasting blood glucose, total cholesterol, HDL and LDL (often computed) cholesterol, triglycerides
Other	Hematocrit, electrocardiogram

4 MECHANISMS OF MEASURING BLOOD PRESSURE

- Conventional Auscultatory Office Blood Pressure
- Automated Office Blood Pressure.
- Home Blood Pressure Monitoring (HBPM)
- Ambulatory Blood Pressure Monitoring

Should have valid **AA** rating from British Hypertensive Society

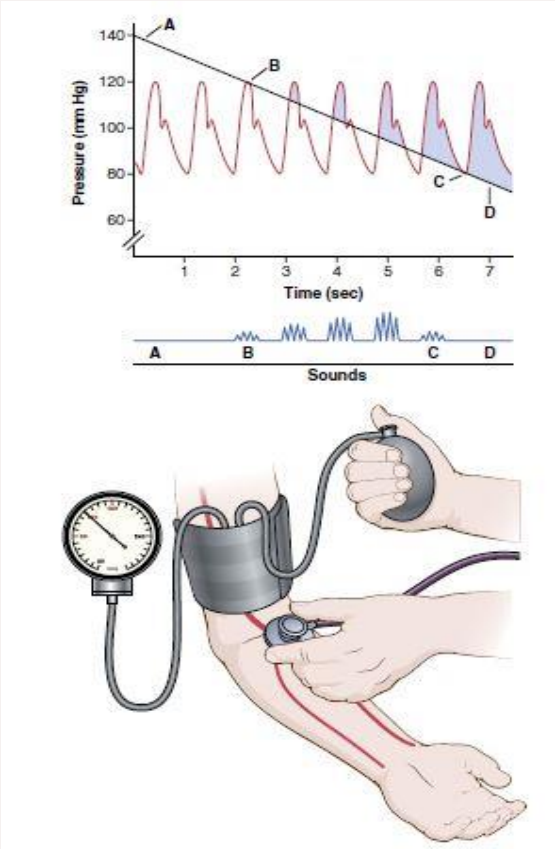
i) Conventional Auscultatory Office Blood Pressure

- **Measured at least twice**
- **After 5 minutes of rest**
- **Patient seated in a chair**
- **Back supported**
- **Arm bare and at heart level**
- **Feet on the floor**

In supine: arm to be raised (mid-atrial level)

Cuff Length: 80%; Width: 40% of arm circumference
Small cuff: Spurious high values

Tobacco and caffeine should be avoided for at **least 30 minutes.**
Measured in both arms and after 5 minutes of standing, (postural fall in BP: autonomic insufficiency).



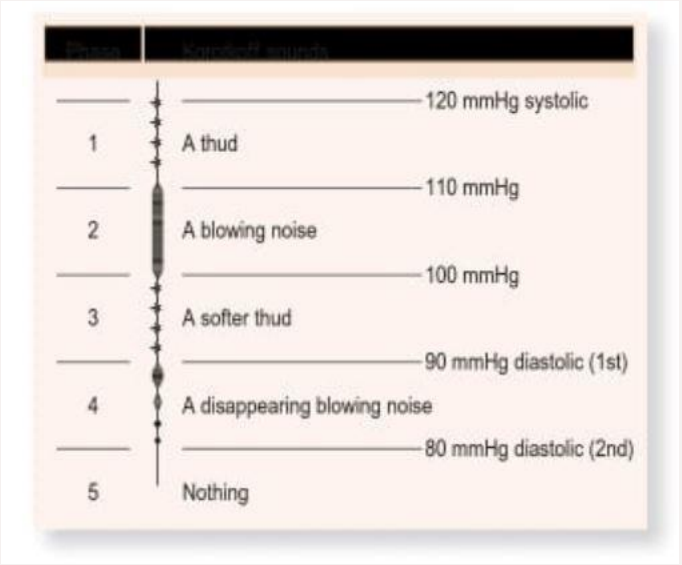
Conventional Auscultatory Office Blood Pressure

Inflate cuff to 30 mmHg over expected systolic
Release at 2mmHg/sec

SBP and DBP: I and V Korotkoff sounds
In conditions with diastolic 'run-off': consider IV Korotkoff sound

Recent regulations prevent use of mercury machines(potential toxicity): use oscillometric/aneroid instead

Auscultatory gap:
Low SBP or high DBP



Classification of office BP

ESH 2018/IGH-IV	Systolic, mmHg	Diastolic, mmHg
Optimal	<120	And <80
Normal	120-129	Or 80-85
High normal	130-139	Or 85-89
Stage 1	140-159	Or 90-99
Stage 2	160-179	Or 100-109
Stage 3	>/= 180	Or >/=110
Isolated systolic	>/=140	And <90

AHA 2017	Systolic, mmHg	Diastolic, mmHg
Normal	<120	And <80
Elevated	120-129	And <80
Stage 1	130-139	Or 80-89
Stage 2	>/=140	>/=90

*Individuals with SBP and DBP in 2 categories should be designated to the higher BP category.

BP indicates blood pressure (based on an average of ≥2 careful readings obtained on ≥2 occasions, as detailed in Section 4); DBP, diastolic blood pressure; and SBP, systolic blood pressure.

ii) Automated Office Blood Pressure.

Recent guidelines endorse AOBP over conventional:

Minimizes white coat reaction
Correlates better with home or awake ambulatory BP
Eliminates digit preference

AOBP is 15/10 mm Hg lower than conventional
office BP: not recommended by IGH IV

HTN if AOBP is 135/85 mm Hg or higher.



iii) Home Blood Pressure Monitoring (HBPM)



Office BP can both overestimate and underestimate actual BP
Involving patients in their own medical care.

Procedure:
Rest for 5 minutes
Seated position
Back supported
Arm supported on a table at heart level
2 readings in the morning and 2 readings in the evening 3-7 days.
Discard first day and calculate the average of remaining days

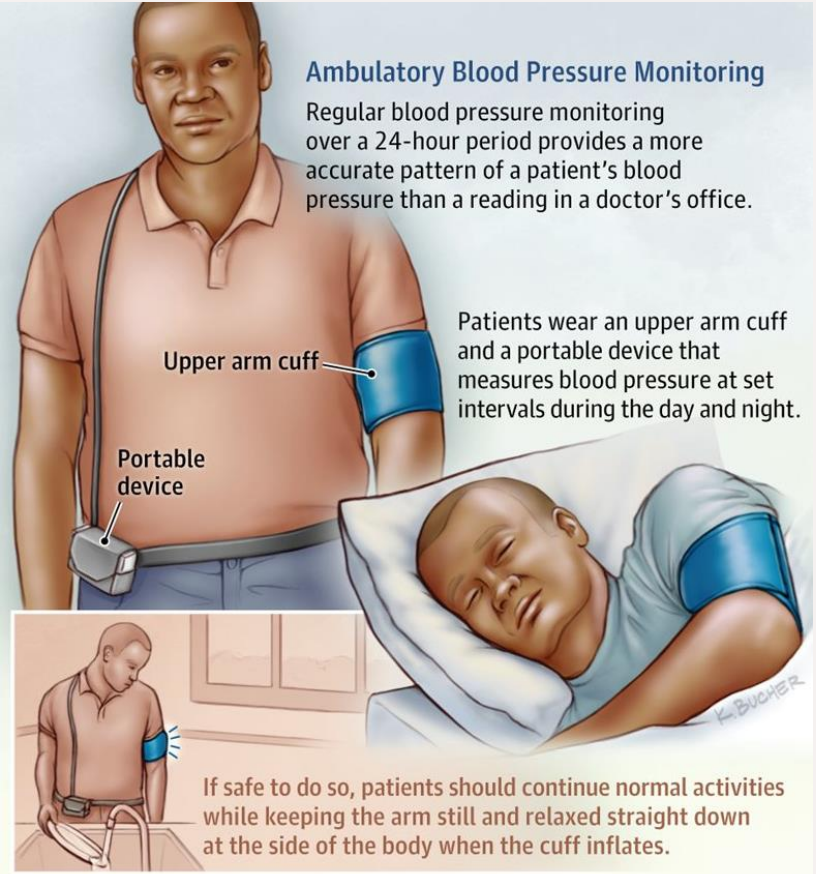
HTN if BP >/= 135/85 mm Hg (ESH 2018)

iv) Ambulatory Blood Pressure Monitoring
"gold standard"

Out-of-office hypertension: (ESH 2018)
- Average daytime BP of >/=135/85 mm Hg or
- nighttime BP >/= 120/70 or
- 24-hour BP >/=130/80 mm Hg

Optimal values:
- Daytime BP less than 130/80,
- nighttime BP less than 110/65, and
- 24-hour BP less than 125/75.

At least two measurements per hour during the patient's waking hours and once at hour at night.

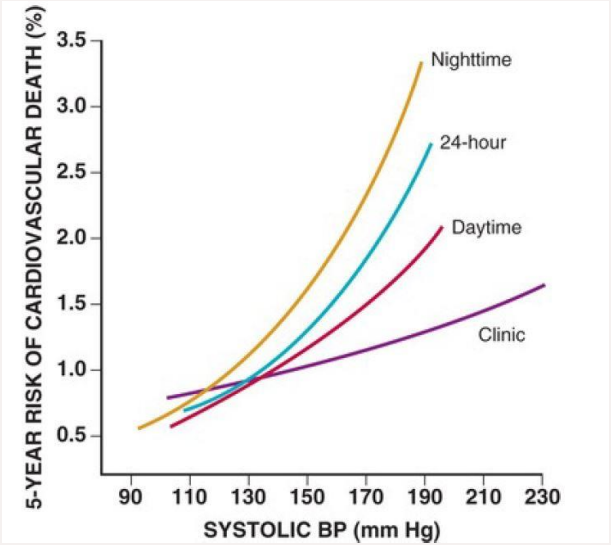


Nocturnal hypertension:

ABPM: only way to detect high BP during sleep

BP normally dips during sleep
Sharp increase on awakening

Absence of dip:
increased hemodynamic load
predicts CVD outcomes



White coat hypertension (15-20%)

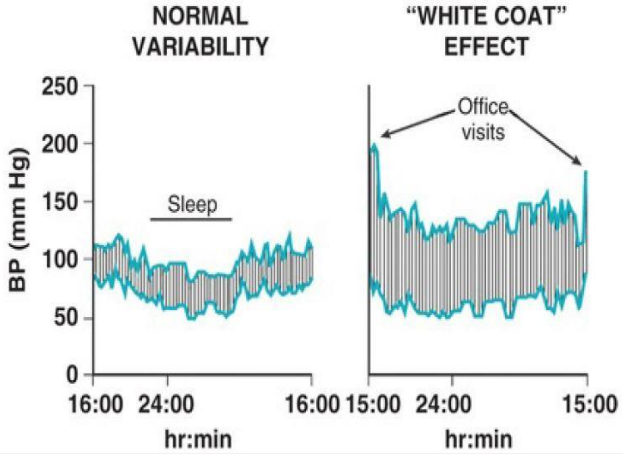
Patients with **elevated office BP** but **normal home and ambulatory BPs**.
Due to **transient adrenergic response** to the measurement of BP only in the OPD.

No target-organ damage

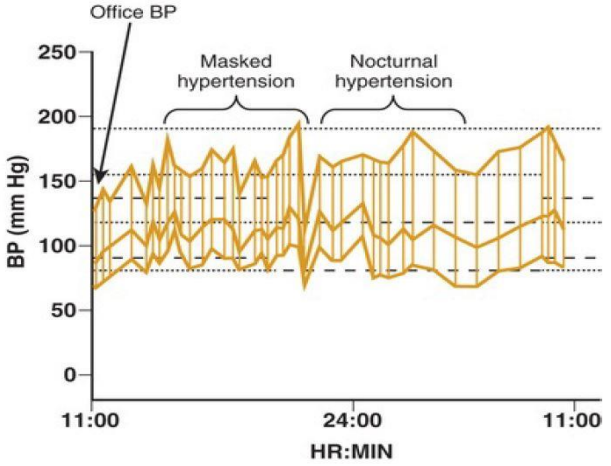
Completely benign / intermediate level of CVD risk

Increases sharply with age

"**White coat aggravation**," a white coat reaction superimposed on a milder out-of-office hypertension
Requires treatment



Masked hypertension



Office readings underestimate out-of-office BP

Definition:

- Daytime BP of >/= 135/85 mmHg or
- Nighttime BP of >/= 120/70 mmHg and
- Conventional office BP lower than 140/90 mm Hg

Increased CVD risk

Sympathetic over activity in daily life caused by job, home stress, tobacco abuse, adrenergic stimulation that disappears when they come to the OPD.

More common in

- African American patients
- Diabetes
- CKD
- Patients being treated with BP medication

Indications for HBPM/24h ABPM: ESH 2018

- To rule out white-coat hypertension: (prevent unnecessary Rx)

Grade I hypertension on office BP measurement
Marked office BP elevation without TOD

- To rule out masked hypertension: (prevent over Rx)

High–normal office BP
Normal office BP in individuals with TOD or at high total CV risk

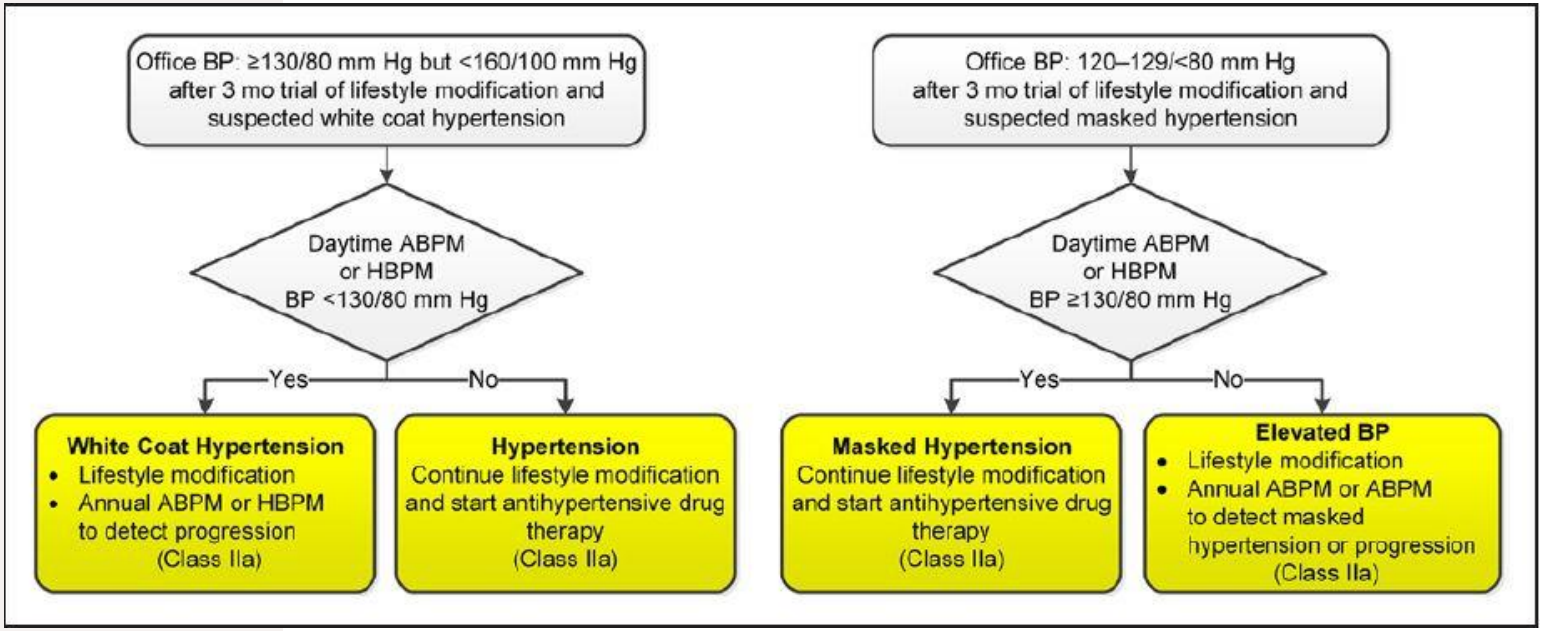
- Resistant hypertension

- Postural and post-prandial hypotension in untreated and treated patients

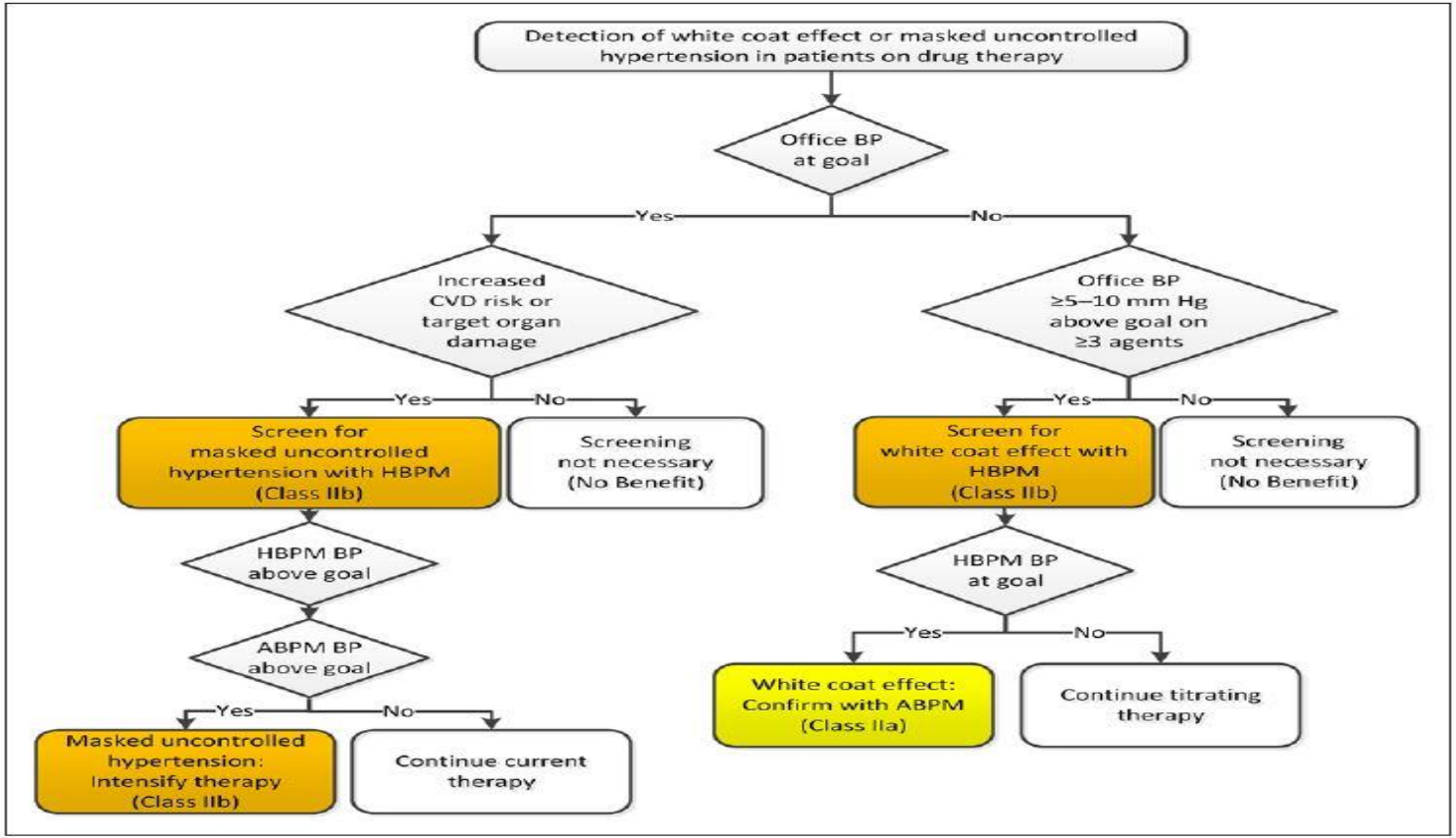
- When there is considerable variability in the office BP

ABPM (and not HBPM)

- Assessment of nocturnal BP and dipping status (sleep apnea, CKD, diabetes, autonomic dysfunction)



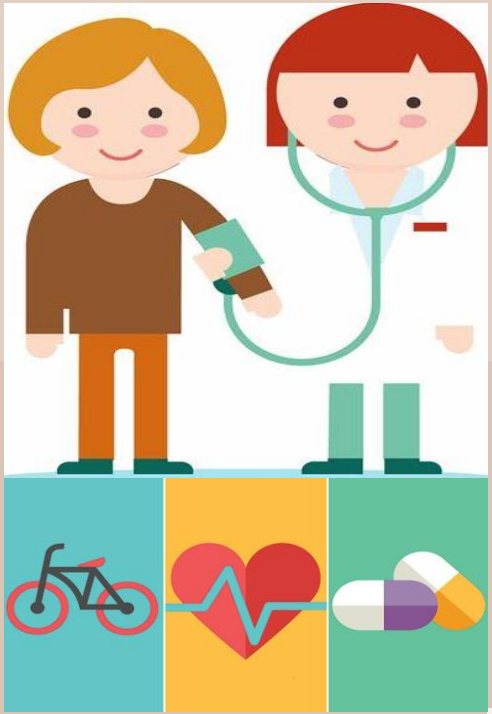
ABPM in patients not receiving drug therapy (AHA 2017)



ABPM in patients receiving drug therapy (AHA 2017)

Treatment

Lifestyle interventions
Pharmacological therapies
Guidelines



LIFESTYLE MODIFICATIONS

Prehypertension: can prevent progression
Hypertension: adjunct to drugs

1) Patient Education

2) Weight loss and physical activity

- Visceral adiposity reduction
- 3-4 sessions per week: 40 minutes per session: moderate to vigorous intensity
- 6.4/3.1 mmHg fall with mean weight reduction of 9.2kg
- Yoga and mediation (IGH)

3) Moderation of alcohol consumption:

- Max allowance: one drink per day in women and 2 in men (1 drink: 14g)

4) Cigarette smoking cessation: (including e-cigarettes)

- Transient pressor response
- Confounded by increase in waist girth

5) DIETARY MODIFICATIONS

Reduction of sodium consumption: <1.5g/d
Avg Indian salt: 13.8g/d; recommendation is 5g
Individual variation in response
Reduction of sodium by 1g/d: reduces SBP by 3-4mmHg
Overly rigorous: Adverse cardiovascular outcomes in DM and CHF

Increased Potassium(3.5-5g) and **Calcium** consumption

DASH Diet (Dietary Approaches to stop Hypertension)
Effective in **high-normal or mild hypertension**.
Rich in fruits, vegetables, low fat dairy, low saturated fats
NaCl<6g
SBP reduction by 5mmHg

Mediterranean Diet: Some studies support

PHARMACOLOGICAL THERAPY

BP reduction has significant Cardiovascular benefit
Individual variation in response
Combinations are used

1) Diuretics: volume loss
Long term: persistent sodium loss and activation of ATP sensitive K channels
Vasodilation and decreased peripheral resistance

2) RAAS blockers:
Vasodilation by reduction of Angitensin 2 action
Reduced Aldosterone: volume reduction

3) Calcium channel blockers:
L-channel blockage: Reduce intracellular calcium: vasodilation

4) Beta blockers:
Decrease cardiac output

5) Direct Vasodilators:
Decrease peripheral resistance

6) α-adrenergic blockers:
Decrease peripheral resistance

7) Sympatholytic agents:
Decrease peripheral resistance

Equivalent BP-lowering effects as monotherapy:
Thiazide diuretic, ACE inhibitors, ARBs, CCBs, Beta-blockers and Alpha 1 blockers

- Benefit proportional to BP reduction
- Intensive BP lowering strategy is better to prevent stroke and MI (SPRINT trial) but S/E are more; So, individualized clinical decision.

DRUG: Thiazide diuretics

MOA: Block Na/Cl symport in early DCT

Recommendation: First line
Chlorthalidone and Indapamide > hydrochlorothiazide (MRFIT trial and HYVET trial)

S/E profile: Hypokalemia, Hyponatremia, Hyperuricemia, Hyperglycemia, Elevate serum lipids
Photosensitive dermatitis; Erectile dysfunction

Preferred in:	Avoided in:
Osteoporosis	Gout Metabolic syndrome Pregnancy

Dosage: Chlorthalidone 12.25-25 OD, hydrochlorothiazide: 25-50 OD

DRUG: Loop diuretics

MOA: Block Na-K-2Cl cotransporter in the thick ascending limb
Most potent diuretics (high ceiling)

Recommendation: Reserved for patients with
Reduced GFR, CHF, edema due to other reasons

S/E profile: Hypotension, Hypokalemia, Hyponatremia, Metabolic alkalosis
Hyperglycemia, Hyperuricemia, Hyperlipidemia, Ototoxicity

Preferred in:	Avoided in:
CKD Symptomatic HF	Combination with ototoxic drugs Gout

Dosage: Furosemide 20-80 BD, Torsemide: 5-10 OD

DRUG: K-sparing diuretics

MOA: Aldosterone antagonists and ENaC channel blockers (late DCT and CD)

Recommendation: Add on drug
Useful in low renin primary hypertension, resistant hypertension

S/E profile:
Spironolactone: Gynecomastia Impotence and menstrual abnormalities
Triamterene: Photosensitivity

Preferred in:	Avoided in:
Primary hyperaldosteronism Liddle's (Amiloride and Triamterene) CHF Resistant hypertension	Acute/Severe renal failure Hyperkalemia Along with RAAS blockers ENaC blockers in CKD with eGFR<45ml/min

Dosage: Spironolactone: 25-100 OD, Eplerenone: 50-100 OD or BD; Amiloride: 5-10 OD or BD

DRUG: ACE inhibitors

MOA: Inhibits conversion of Angiotensin I to Angiotensin II
Increased Substance P: vasodilator; reduce sympathetic nervous system activity

Recommendation: First line

S/E profile: Hyperkalemia, taste disturbances
Increased Bradykinin: cough(15%) and angioedema(<1%)
Captopril: First dose effect, leucopenia

Preferred in:	Avoided in:
CHF (inhibits cardiac remodeling) LVH CKD (reduced dose: monitor K) In diabetic patients (improve insulin action)*	Pregnancy B/l renal artery stenosis U/l RAS in a solitary kidney With NSAIDs: may precipitate ARF African-Americans (low renin HTN) Combination with other RAAS blockers

Dosage: Ramipril 2.5-20 OD or BD; Enalapril: 5-40 OD or BD

*Recent RCTs do not agree: ALLHAT

DRUG: Angiotensin receptor blockers

MOA: AT1 receptor blockade

Recommendation: First line
Losartan: uricosuric effect
Telmisartan: PPAR-γ agonist activity (preferred in diabetics)

S/E profile: Hyperkalemia (especially in CKD, DM, RTA type 4)

Preferred in:	Avoided in:
Same as ACE –	Combination with ACE inhibitor
Angioedema with ACE inhibitor	Same as ACE –
Gout: Losartan	

Dosage: Telmisartan: 20-80 OD; Olmesartan: 20-40 OD; Losartan: 50-100 OD or BD

DRUG: Direct Renin Inhibitors

MOA: Prevents conversion of prorenin to renin
More complete RAAS inhibition
Monotherapy as effective as ARB and ACE inhibitors

Recommendation:
Add on drug

S/E profile: Similar to ACE – and ARB
Increased prorenin: potentially harmful (fibrosis)

Preferred in:	Avoided in:
Similar to ACE – and ARB	Combination with ACE inhibitor or ARB

Dosage: Aliskiren: 150-300 OD

DRUG: Calcium channel blockers

MOA: L-type channel blockage
Vasodilation (DHP>diltiazem>verapamil)
Some diuretic action (afferent arteriole dilation)

Recommendation: First line

S/E profile: Gingival hyperplasia (rare but serious), GERD
Some DHPs: Flushing, headache and edema (potent arterial dilators): reflex tachycardia
Verapamil: constipation (dose-dependent), impaired cardiac conduction (bradycardia)

Preferred in:	Avoided in:
Strained economy Concurrent NSAID use Safe in PVD, asthma Pregnancy	CKD (as monotherapy: less reno-protective) Verapamil and Diltiazem: CHF, AV block With digoxin, beta-blockers or central sympatholytic agents Nifedepine: angina, MI, arrhythmias

Dosage: Amlodipine 2.5-10 OD, Verapamil: 120-360 TID, Diltiazem: 120-360 OD

DRUG: Beta-blockers

MOA: β1 receptor blockade: reduced heart rate and contractility: reduced cardiac output
Inhibit renin release

Recommendation: Add on drug
Newer beta=blockers are class 1 in IGH-IV

S/E profile: Increased risk for diabetes and hypoglycemic unawareness
Fatigability
Impaired cardiac conduction

Preferred in:	Avoided in:
Tachycardia Concomitant heart disease including CHF Pregnancy HFfEF (Carvedilol, Bisoprolol and Metoprolol)	Asthma and COPD AV Block PVD DM and metabolic syndrome

Dosage: Atenolol: 25-100 BD; Metoprolol 50-200 OD; Carvedilol: 12.5-50 BD; Propranolol 80-160 OD/BD; Nebivolol 5-40 OD

DRUG: Direct Vasodilators

MOA: open ATP sensitive K channels; Nitroprusside increases NO

Recommendation: Add on drug
Preferably with loop diuretic (prevents fluid retention) and beta blocker (prevents reflex tachycardia

S/E profile:
Reflex sympathetic activation: tachycardia ,RAAS activation
Hydralazine: lupus like syndrome
Minoxidil: Hypertrichosis and pericardial effusion

Preferred in:
Pregnancy
Along with nitrates in HF
Advanced CKD with severe HTN (with loop diuretic and betablocker)<MHD

Dosage: Hydralazine 100-200 OD or BD

DRUG: α-adrenergic blockers

MOA:
Post synaptic α-receptor blockage: vasodilation: decreases peripheral resistance

Recommendation: Add on drug
Increases insulin sensitivity (by increasing blood flow to skeletal muscles)

S/E profile: Fluid retention, orthostatic hypotension
Tachyphylaxis: unmasks/exacerbates HF

Preferred in:
BPH (dilate urethral smooth muscle): tamsulosin
Pheochromocytoma: Phenoxybenzamine

Dosage: Prazocin 2-20 BD or TID

DRUG: Sympatholytic agents

MOA:
Post synaptic α2 receptors in CNS: reduced central sympathetic outflow
Pre-synaptic α2 receptors: feedback inhibition of NE from peripheral sympathetic nerve terminals

Recommendation:
Add on drug for very difficult hypertension

S/E profile: Reduced QOL
Somnolence, dry mouth, rebound HTN (on withdrawal)

Preferred in:
Autonomic neuropathy
Hypertensive urgencies

Dosage: 0.1-0.8 BD

NON-PHARMACOLOGICAL THERAPY

Inhibit sympathetic drive
Decreasing renin release

Renal Denervation

Percutaneous catheter-based radiofrequency ablation of the renal nerves
Minimally invasive
Simplicity HTN-3 Trial:
No benefit

Carotid Baroreflex Activation Therapy

Electrode implantation on both the right and left carotid arteries
Phase III Rheos Pivotal Trial:
Equivocal results

BLOOD PRESSURE GOALS

Half-standard doses ~20% less effect than standard doses.
Use low-dose combinations

ACCORD trial: No point of intensive BP lowering in DM

Older individuals:
In isolated systolic hypertension, further fall in DBP on treatment okay. (J curve)
Less aggressive target level of control (e.g., 130–150 mmHg).

Table 23. BP Thresholds for and Goals of Pharmacological Therapy in Patients With Hypertension According to Clinical Conditions

Clinical Condition(s)	BP Threshold, mm Hg	BP Goal, mm Hg
General		
Clinical CVD or 10-year ASCVD risk ≥10%	≥130/80	<130/80
No clinical CVD and 10-year ASCVD risk <10%	≥140/90	<130/80
Older persons (≥65 years of age; noninstitutionalized, ambulatory, community-living adults)	≥130 (SBP)	<130 (SBP)
Specific comorbidities		
Diabetes mellitus	≥130/80	<130/80
Chronic kidney disease	≥130/80	<130/80
Chronic kidney disease after renal transplantation	≥130/80	<130/80
Heart failure	≥130/80	<130/80
Stable ischemic heart disease	≥130/80	<130/80
Secondary stroke prevention	≥140/90	<130/80
Peripheral artery disease	≥130/80	<130/80

ASCVD indicates atherosclerotic cardiovascular disease; BP, blood pressure; CVD, cardiovascular disease; and SBP, systolic blood pressure.

AHA 2017

Table 23 Office blood pressure treatment target range

Age group	Office SBP treatment target ranges (mmHg)					Office DBP treatment target range (mmHg)
	Hypertension	+ Diabetes	+ CKD	+ CAD	+ Stroke[a]/TIA	
18 - 65 years	**Target to 130** *or lower if tolerated* Not <120	**Target to 130** *or lower if tolerated* Not <120	**Target to <140 to 130** *if tolerated*	**Target to 130** *or lower if tolerated* Not <120	**Target to 130** *or lower if tolerated* Not <120	70–79
65 - 79 years[b]	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	70–79
≥80 years[b]	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	**Target to 130-139** *if tolerated*	70–79
Office DBP treatment target range (mmHg)	70–79	70–79	70–79	70–79	70–79	

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CAD = coronary artery disease; CKD = chronic kidney disease (includes diabetic and non-diabetic CKD); DBP = diastolic blood pressure; SBP = systolic blood pressure; TIA = transient ischaemic attack.

[a]Refers to patients with previous stroke and does not refer to blood pressure targets immediately after acute stroke.

[b]Treatment decisions and blood pressure targets may need to be modified in older patients who are frail and independent.

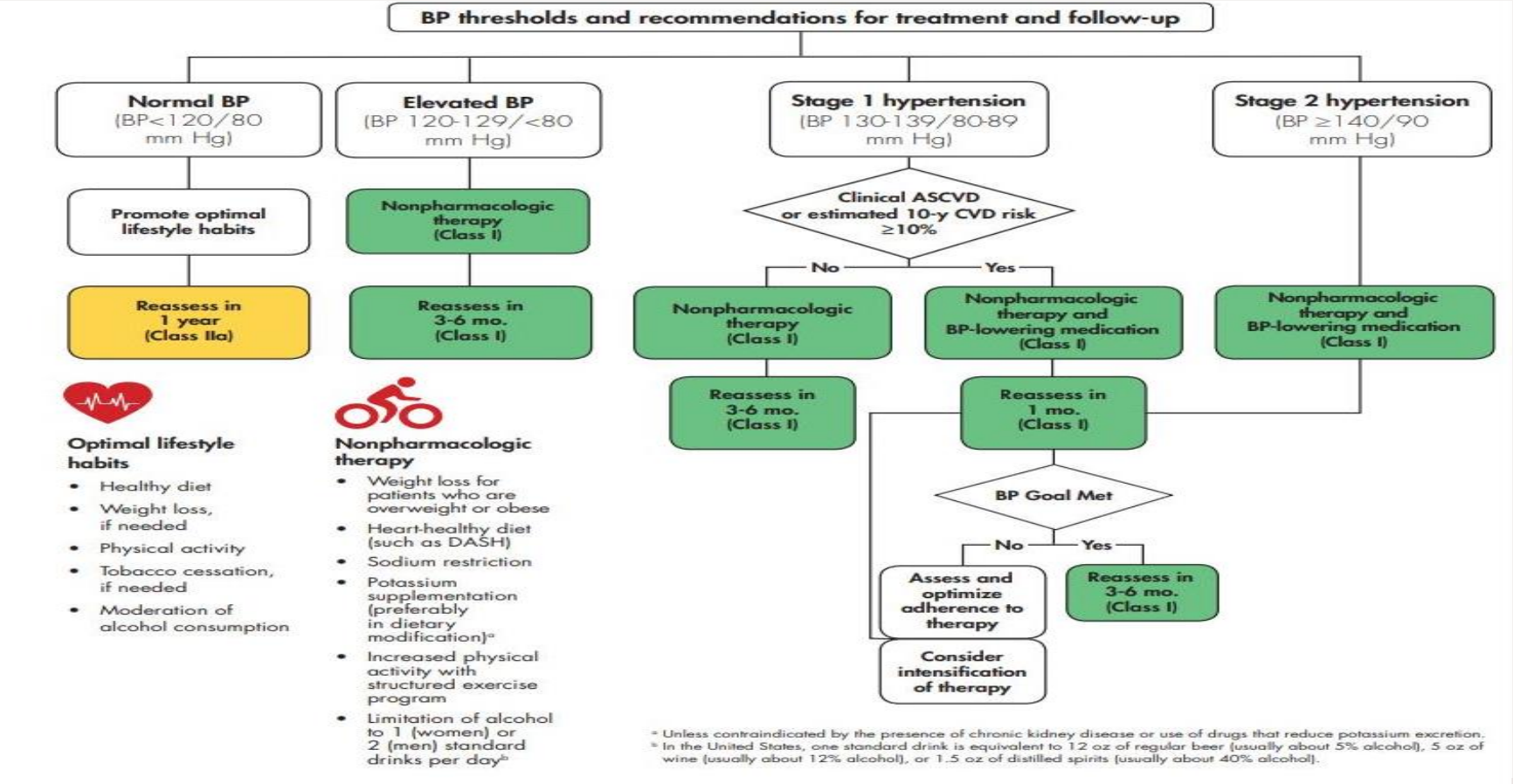
MANAGEMENT GUIDELINES

A brief comparison between AHA 2017 and ESH 2018 guidelines





AHA Treatment recommendations



Current Age *

Age must be between 20-79

Sex *

Male | Female

Race *

White | African American | Other

Systolic Blood Pressure (mm Hg) *

Value must be between 90-200

Diastolic Blood Pressure (mm Hg)

Value must be between 60-130

Total Cholesterol (mg/dL) *

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

Value must be between 20 - 100

LDL Cholesterol (mg/dL)

Value must be between 30-300

History of Diabetes? *

Yes | No

Smoker? *

Current | Former | Never

On Hypertension Treatment? *

Yes | No

On a Statin?

Yes | No

On Aspirin Therapy?

Yes | No

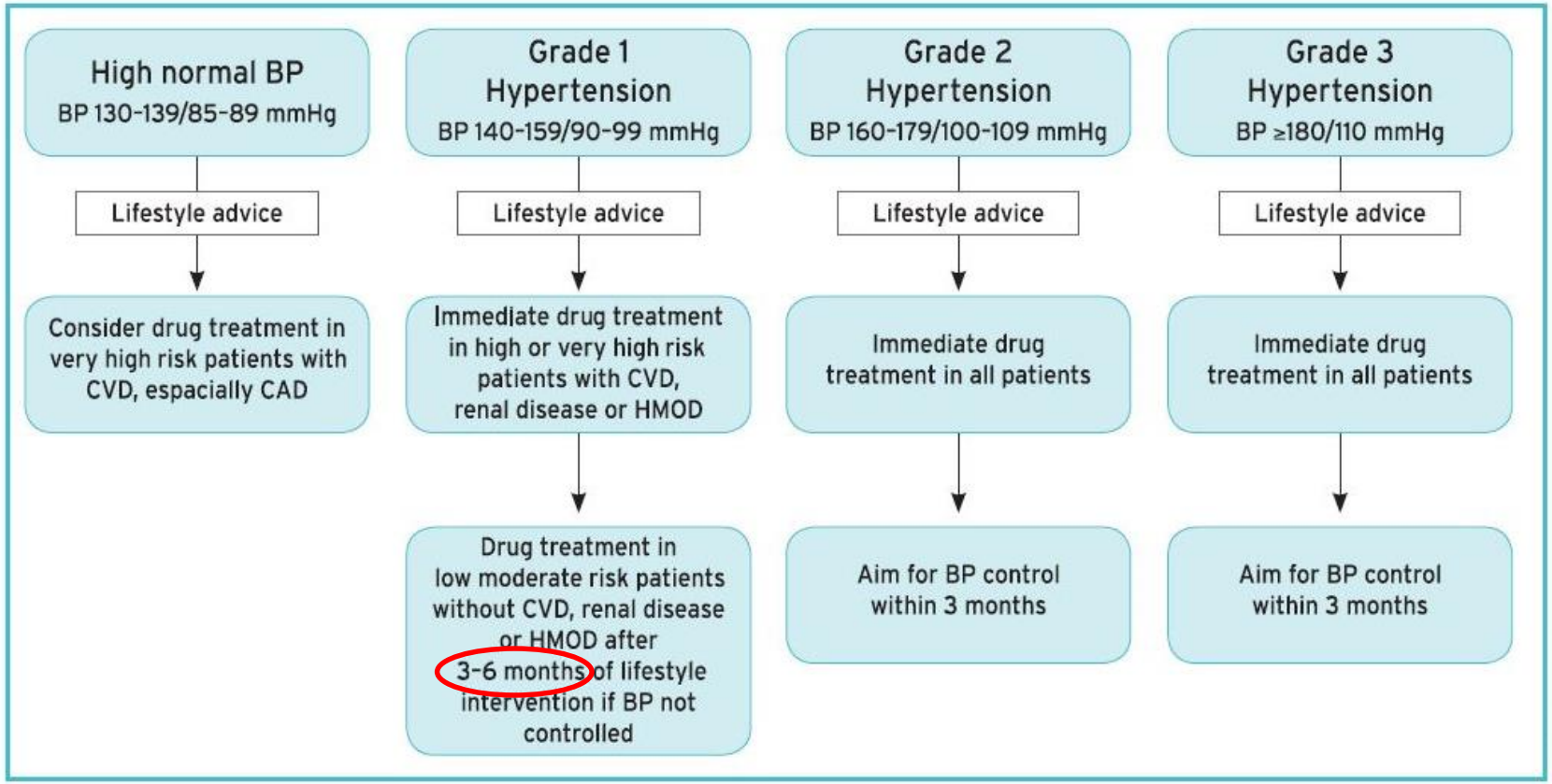
ASH guidelines for co-morbidities

	Nil and DM	SIHD	CKD	HFrEF	Old Stroke
Step 1	A/C/D	A/B	A* A/C/D	B/A/D/MRA	A/C/D If: >140/90
Step 2	Double/Triple combination	Add C/D/MRA Angina C	Double/Triple combination	Double/Triple combination	Double/Triple combination
Step 3	TC + others	TC + others	TC + others		
Target	130/80	130/80	130/80	130/80	130/80

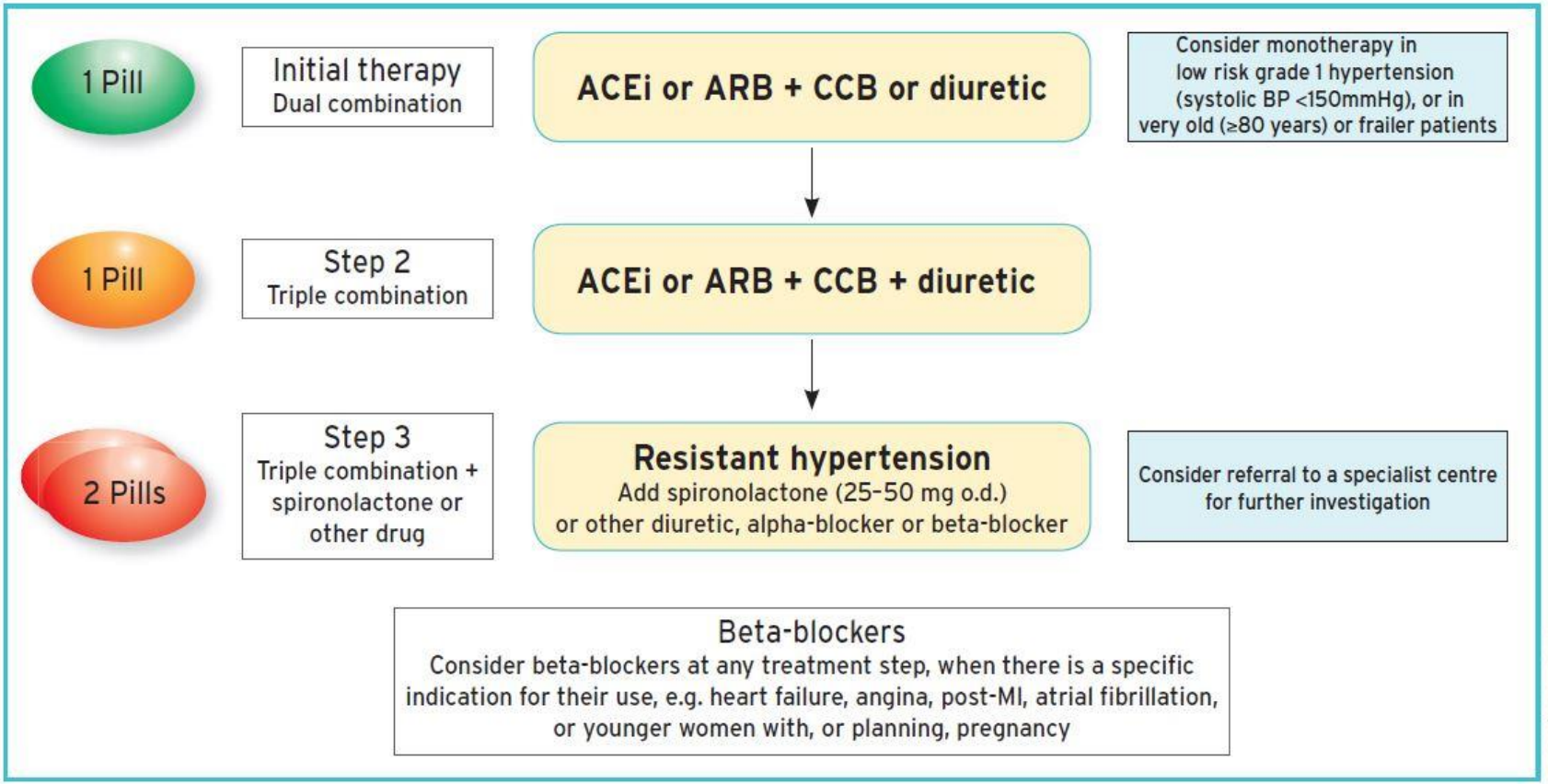
A: ACE inhibitors/ARBs
B: Beta-blockers
C: CCB
D: Thiazide diuretics
MRA: MC receptor antagonist

*: Albuminuria >300mg/d
**: non DHP CCB

ESH Treatment recommendations



Systematic COronary Risk Evaluation (SCORE) system to determine cardiovascular risk

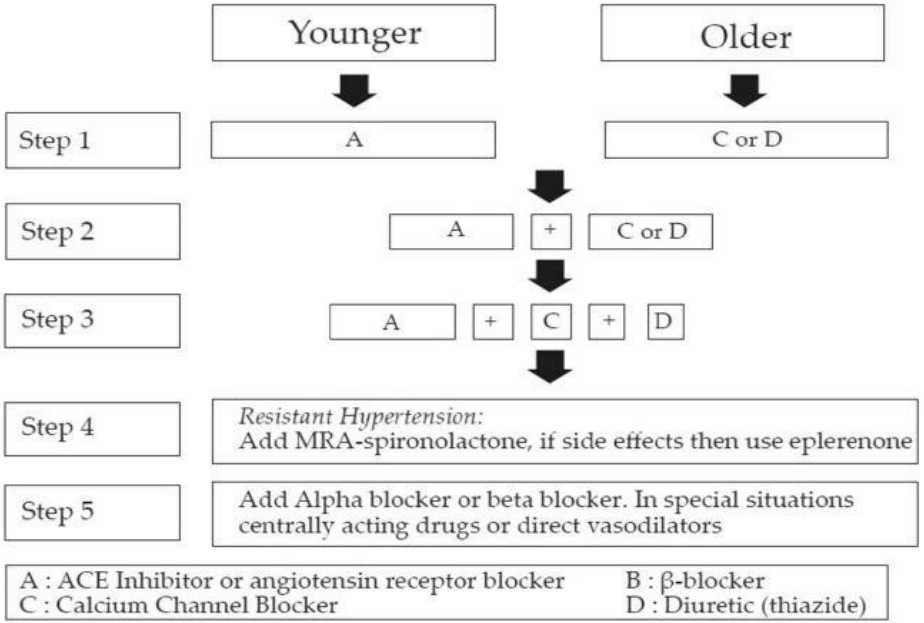

1 Pill
Initial therapy
Dual combination
ACEi or ARB + CCB or diuretic
Consider monotherapy in low risk grade 1 hypertension (systolic BP <150mmHg), or in very old (≥80 years) or frailer patients
1 Pill
Step 2
Triple combination
ACEi or ARB + CCB + diuretic
2 Pills
Step 3
Triple combination + spironolactone or other drug
Resistant hypertension
Add spironolactone (25–50 mg o.d.) or other diuretic, alpha-blocker or beta-blocker
Consider referral to a specialist centre for further investigation
Beta-blockers
Consider beta-blockers at any treatment step, when there is a specific indication for their use, e.g. heart failure, angina, post-MI, atrial fibrillation, or younger women with, or planning, pregnancy
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ESH guidelines for co-morbidities

A: ACE inhibitors/ARBs
B: Beta-blockers
C: CCB
D: Thiazide diuretics
MRA: MC receptor antagonist

	DM	CAD	CKD	HFrEF	AF
Step 1	A+C/D or C+A/D	A+B/C or C+D/B or B+D (SBP>130)	A+C/D*	A+B+D	A+B/C**or B+C
Step 2	Triple combination	Triple combination	Triple combination	A+B+D and MRA	A+B+C/D B+C+D
Step 3	TC + others	TC + others	TC + others		
Target	130/80	130/80	140/80		

Consider monotherapy in low risk grade 1 HTN if SBP is <150mmHg and in older >80y patients
*: loop diuretic if eGFR< 30ml/min or edema
**: non DHP CCB



If BP goal is achieved:
Follow-up 3-6 months

Not achieved:
Ensure compliance
Follow algorithm
Reinforce lifestyle measures

Significant S/E:
Substitute drugs from other classes

IGH-IV approach to pharmacotherapy

Resistant hypertension(13%)

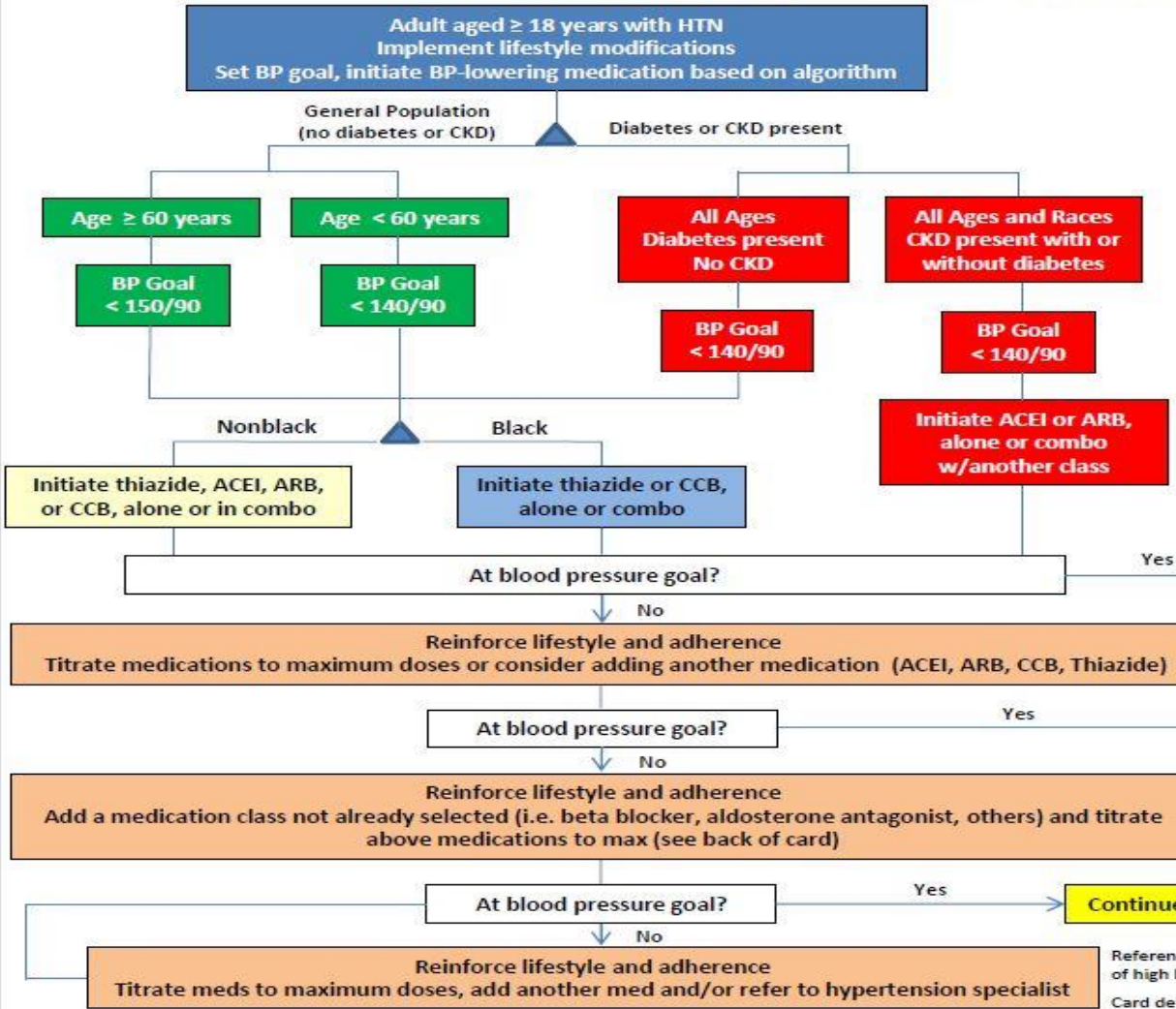
BP **uncontrolled with three** or **controlled with at least four drugs** (including a **diuretic**)

Distinguish from **Pseudo-Resistant hypertension**
Osler maneuver: sclerosed brachial artery

Causes:
- Chronic kidney disease
- Primary aldosteronism
- Other secondary hypertension
- Difficult primary hypertension

Add-on drugs:
Mineralocorticoid receptor antagonist
Vasodilating beta blocker
Loop diuretics in CKD

JNC 8 Hypertension Guideline Algorithm



Initial Drugs of Choice for Hypertension

- ACE inhibitor (ACEI)
- Angiotensin receptor blocker (ARB)
- Thiazide diuretic
- Calcium channel blocker (CCB)

Strategy	Description
A	Start one drug, titrate to maximum dose, and then add a second drug.
B	Start one drug, then add a second drug before achieving max dose of first
C	Begin 2 drugs at same time, as separate pills or combination pill. **Initial combination therapy is recommended if BP is greater than 20/10mm Hg above goal**

Lifestyle changes:

- Smoking Cessation
- Control blood glucose and lipids
- Diet
  - ✓Eat healthy (i.e., DASH diet)
  - ✓Moderate alcohol consumption
  - ✓Reduce sodium intake to no more than 2,400 mg/day
- Physical activity
  - ✓Moderate-to-vigorous activity 3-4 days a week averaging 40 min per session.

Reference: James PA, Ortiz E, et al. 2014 evidence-based guideline for the management of high blood pressure in adults: (JNC8). JAMA. 2014 Feb 5;311(5):507-20

Card developed by Cole Glenn, Pharm.D. & James L Taylor, Pharm.D.

Hypertensive Emergency and Urgency

BP>180/120
Rate of rise > Absolute value



Hypertensive urgency:
Severe uncontrolled hypertension **without evidence of acute target organ damage**
Managed with oral medications

Hypertensive emergency:
Failure of autoregulation in major organ systems
Associated with **new/worsening target organ damage**:
1. Encephalopathy
2. Retinopathy: (arteriolar spasm, hemorrhages, exudates, and papilledema)
3. Cerebral haemorrhage
4. Acute kidney injury: preoteinuria
5. Aortic dissection
6. MAHA and DIC

Malignant hypertension*:*
Abrupt increase of BP in a patient with underlying hypertensio or sudden onset of hypertension in a previously normotensive individual

Hypertensive encephalopathy:

BP rises beyond autoregulated zone
C/F:
Headache
Nausea and projectile vomiting
Altered mental status
Untreated, rapid progression to stupor coma and death

CT/MRI : cerebral edema
Posterior brain regions(vertebral arteries)
Initially reversible

Reduction by
10% : 1st hour and 15% over next 12 hours (but > 160/110)

Overly aggressive therapy: cerebral ischemia or infarction

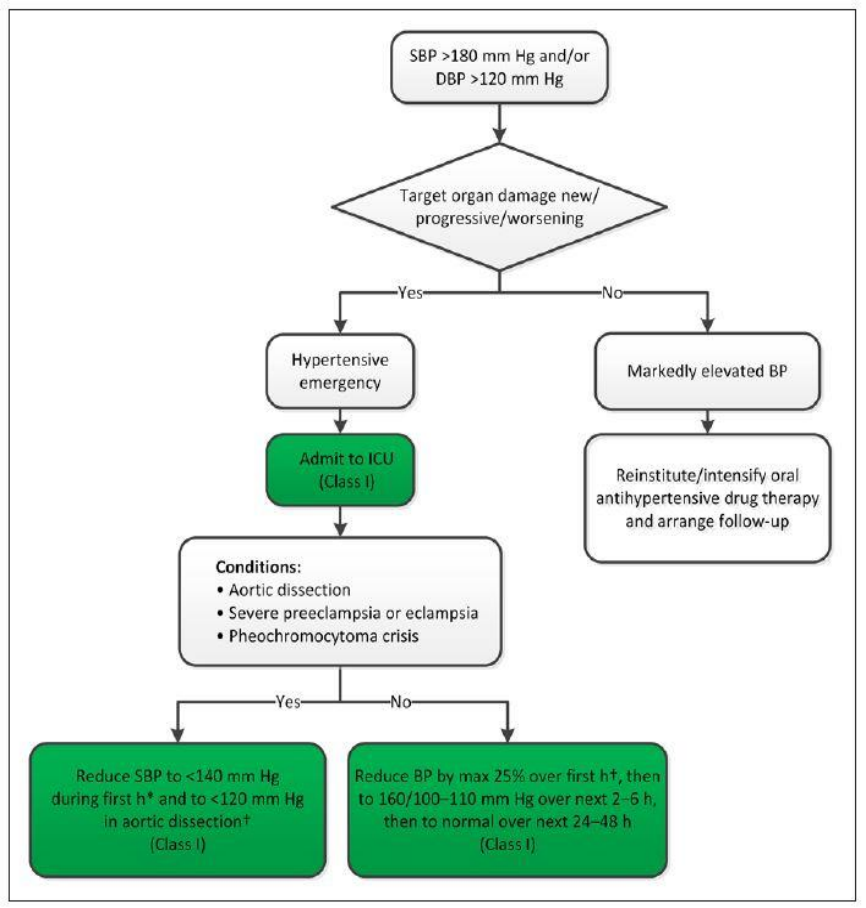
Tests:

- Fundoscopy
- ECG
- Hb, Plt and fibrinogen
- Creatinine, SE, LDH
- Urine routine

First line: (ESH 2018):

- Labetalol: 0.5 mg/kg bolus and 2-4 mg/min iv till BP goal reached f/b 5-20mg/h
- Nicardipine: 5-15mg/h iv infusion

Other drugs: Metoprolol, NTG, Nitoprusside, Clonidine



Sublingual nifedepine is CI

References:

Harrison’s principles of Internal Medicine

Braunwald’s Heart disease: A text book of cardiovascular medicine

2018 ESC/ESH Guidelines for thee management of arterial Hypertension

Indian Guidelines on Hypertension –IV (2019)

2017 AHA Guideline for Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults